

Urban Botany

A Cross Platform Application for the Care of Houseplants in Urban Dwellings

By

Dexter Rowland

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Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

Dexter Rowland

Date: 15/01/2026

Signed (apply signature below)

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Abstract

Maintaining healthy houseplants was identified as a common problem within urban environments through lack of consistency, motivation, and access to information. This project draws on research that humans have an innate connection to nature, nurturing improved emotional regulation, and that gamification mechanisms when properly integrated can improve consistency, motivation, and positive behavioural changes.

A full-stack web application “Urban Botany” was created to support plant care through personalised care routines as well as personalisation and reward-based gamification mechanisms to drive engagement. The system integrates species identification, environmental modelling, and task scheduling.

Evaluation was undertaken through user testing, think-aloud exercise indicated that users engaged positively with the application and benefited from it. Within this process critical feedback was provided regarding user interface design and functionality, which has been addressed appropriately.

The project demonstrates the feasibility of applying gamification mechanisms to positive behaviour modification. While the domain of the project resides in urban houseplant care. The fundamental idea of applying such methodologies to undesirable habitual tasks could be expanded to an extensive list of challenges.

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1. Introduction

This project was motivated by the challenges associated with understanding and maintaining appropriate care routines for houseplants. Many individuals lack the knowledge required to support plant health effectively, particularly within urban environments.

1.1 Problem Statement

Urban areas are constantly expanding[1] [2], with more young people becoming isolated from nature. Considering the innate attraction that humans have to nature[3][4], it is no surprise that these urban dwellers are trying to draw nature into their homes through the care of house plants[5].

However, among the busy lives of these young people comes the struggles of knowing how and when to care for these plants, or how they can expand upon their hobby.

This project addresses gaps in accessible plant care knowledge and user adherence to care routines through the development of a user-centred application.

1.2 Aims

Through the development of this project, I aim to design and implement a mobile application that empowers urban residents to care for plants reliably, reinforce learning about plants, and improve commitment to plant care.

The application should:

1. Provide a user-centric, personalised environment to monitor and check plant care schedules.
2. Inform users promptly when certain care actions need to be undertaken.
3. Integrate AI technology to identify plant species, adding them to the user's environment and care schedule.
4. Implement features to engage users, improving positive habits.

1.3 Objectives

To achieve these goals, I will be focusing on meeting the following objectives:

1. Investigate urban plant-care habits, abstracting aspects of care into a suitable format for storage in a database.
2. Design and create a secure, efficient, populated database for the storage of user data and plant data.
3. Create an application with intuitive mobile UX/UI for plant care scheduling and tracking.
4. Implement features to remind users to undertake care tasks.
5. Update care schedules through user input when a task is undertaken.
6. Create algorithms to adjust routines dependant on external factors.
7. Create and deploy an AI model capable of identifying plant species from image input.
8. Test usability and measure engagement with a pilot group.

1.4 Success Criteria

To ensure I have delivered on the previously mentioned aims and objectives, I will be implementing success criteria in the form of user stories within a GitHub repository. User stories will be abstracted into further checkboxes and added to a kanban board to monitor progress and ensure quality of work.

2 Background and Context

2.1 Background

Urbanisation has significantly reduced human connection to nature. In response, houseplants have started to make a resurgence in domestic and commercial spaces. While the demand to reconnect with nature through urban integration, a lack of knowledge and persistence to provide nurture remains.

Plants rely on a range of factors such as light exposure, water intake, soil type, humidity and temperature. While existing applications try to provide access to knowledge and care information, many applications rely on generic information, not accounting for environmental conditions or retaining user engagement. This lack of overlap reveals a gap in the market for data driven personalised recommendations and an accessible method for improving engagement.

2.2 Summary of Relevant Published Work (Literature Review)

2.2.1 Nature is Nurture:

Lack of access to open green spaces negatively impacts mental health[6]. With changes in ecosystems as a result of human interference, we are becoming more detached from nature[4]. These adverse changes in the environment are detrimental to mental health in humans. Urban expansion has significantly reduced access to natural environments, contributing to decreased biodiversity and reduced human interaction with nature. As stated in Biophilia, “the urge to affiliate with other forms of life is to some degree innate”[4]. E. O. Wilson writes about how it is in our nature to care for life, to want to master it, that “jungles and grasslands are the logical destinations, and towns and farmlands are the labyrinths that people have imposed between”, proving our natural draw to lands filled with life. E. O. Wilson coins this relationship “Biophilia”.

Houseplants aren’t just linked to mental health, but also to physical health [7]. Home interiors are microbiome incubators, with plants acting as filters, improving air quality and metabolising human respiratory products. This study reinforces E. O. Wilson’s concept of biophilia on a new level. Tying

biophilia not just to an affiliation but also to physical health. The relationship between humans and nature covers multiple aspects of health and quality of life.

According to the UN, 68% of the world population will be living in urban areas by 2050[8]. With the current projection of human population growth rates of 9.8 billion in 2050 [9], this could be over 6.6 billion people living in urban areas.

People are turning to houseplants to incorporate biophilic design in homes. However, the climates within high-rises, apartments, flats, detached, semi-detached, bungalows and other home types aren't always embodying of the landscapes in which these plants thrive. Many new houseplant enthusiasts don't know how to care for these organisms. With each plant requiring different care needs it can be hard to provide sufficient care without reminding, monitoring or informing.

Despite increasing interest in houseplants, users often lack the knowledge required to determine appropriate care conditions, including light, watering frequency, and placement.

With so many questions, with answers unique to specific species, it can become overwhelming. Care schedules have been created on calendars in the past, but what pull is there towards checking a calendar or gaining insight? How can one be reminded to check this log? This introduces our second problem, forming habits.

2.2.2 The Problem of Perseverance:

Even if one intends to form a positive habit, there is no guarantee that a habit will form from good intention[10]. The study by U. N. Danner et al hypothesises (reinforced by Triandis [11]) that as a behaviour is more frequently executed, actions become less guided by intention and habit takes over as the driving factor. To build new positive habits and improve perseverance, methods to increase repetition of behaviour in a consistent context must be implemented [12]. Many web applications utilise reward-based intermittent reinforcement. A way of providing small prizes at random intervals to keep users engaged [13]. Behavioural reinforcement mechanisms are widely used in digital platforms [14] and can be adapted to support positive habit formation as shown by Duolingo[15].

To ethically integrate these methodologies, a code of ethics needs to be deployed within the planning stages to ensure compliance throughout the development of the artefact. Companies such

as Instagram, Facebook and TikTok rely on these techniques. Authors J. R. Saura et al [14] identified 14 performance measurements linked to ethical design in social media to be explored later within the ethics section.

Use engagement must be balanced to encourage habit formation without promoting excessive or compulsive use. An ideal engagement target should be defined to assist with the design of these engagement mechanics. As ethical designers the aim is not to build addiction mechanisms keeping users constantly engaged. The aim as previously outlined is to form positive habits through perseverance. Plant care tasks need to only be undertaken once per day and not require constant engagement, as such, methodologies should be designed to meet these criteria, not exceed them.

2.2.3 Integration of Artificial Intelligence:

With the recent demand for systems to contain artificial intelligence agents, it appears prudent to implement such features within the project space. Artificial intelligence techniques are incorporated to enable plant species identification and personalised care recommendations.

As defined in the initial aims, identification is one potential implementation of an AI agent, utilising physical sensors such as cameras, to impart changes to a user's database and add a new plant species to their account through database queries (actuators). Additional implementations of AI include the curation of custom care guides tailored to user profiles.

With artificial intelligence being an emerging technology consistently in the public eye, there are papers perpetually being published on ethical concern. Technology is being developed at speeds that ethics, law and policy cannot keep up. To navigate this issue within the confines of this report, a framework for the ethical implementation of artificial intelligence agents will be developed and implemented. The framework should form a foundational structure, on which future systems can be formed, to ensure ethical compliance and risk mitigation.

Table 1: Key Literature Reviewed

Literature	Topic	Benefits	Limitations
Biophilia	The Human Connection to Nature.	Highlights human's innate connection to nature.	Mostly refers to ants throughout the literature.
Houseplants as home health monitors	Physical health improvements from plants	Highlights the importance of plants for physical health.	Mostly targeted at urban architecture.
Habit vs. intention in the prediction of future behaviour	The importance of repetition and intention relating to habit formation.	Performs 3 studies and analyses the results.	Intention is a psychological, non-quantifiable metric.
Ethical design in social media	Ethical design of behaviour modification practices	Identifies and analyses 14 indicators for addiction.	Scientific literature the article is based on is "insufficiently advanced".

2.3 Competitor Analysis and Related work

Digital care schedules are no new concept. Many other products exist on the market to help people provide more sufficient care towards their houseplants. Within this section, a few of these products are analysed after first-hand usage and experimentation. The elements of these products are defined and critically analysed in a non-biased approach. Their strengths are highlighted and praised, while their flaws are analysed. To summarise this section, I will propose the targets to be met within my project to create a unique product.

2.3.1 Plant Parent:

Plant parent is a commercially available mobile application for plant care guidance, disease diagnosis, and scheduled reminders. Its primary strength lies in the breadth of its dataset, supporting thousands of plant species alongside automated disease and ailment diagnostic functionality. This provides the user with a comprehensive tool to reduce error from external factors and self-imposed limitations.

However, despite the functional strength of the application, there are limitations in the user experience design. There is a distinct lack of navigational clarity, and limited customisation restricts user engagement. Additionally, environmental recommendations are based on general pre-existing data. Enforcing assumptions rather than user-specific conditions, reducing accuracy in real-world translation.

These limitations highlight key areas for improvement within this project, specifically the integration of personalisation for both engagement and improved accuracy.

2.3.2 Willow:

Willow stands out through the integration of physical hardware sensors to monitor environmental conditions such as soil moisture, light levels, humidity, and temperature. This data is fed back to the user and built into its plant care model. Providing highly accurate, real-time feedback, significantly reducing the risk of improper care.

However, this reliance on continuous monitoring introduces potential drawbacks. Frequent updates and notifications may cause excessive user engagement, raising the previously mentioned concern about dependency and over-reliance instead of building positive routine. Constant monitoring may also issue the user with an abundance of tasks such as re-location of a plant throughout the day.

While willow represents an interesting product within this space, demonstrating clear benefits of real-time environmental data. This project will explore based approximation, using user input. Aiming to balance accuracy with more accessible, lower cost methods.

2.3.3 Planta:

Planta offers an advanced plant care system with strong improvements on location refined recommendations. Its use of environmental groupings (“sites”) and consideration of environmental variables including light exposure, pot characteristics, and climate enables more refined care routines in comparison to similar applications.

A key strength of the product is the ability to account for multiple factors within its recommendations, increasing model accuracy and subsequent recommendations. The application also features shared responsibilities, enhancing user engagement and introducing a reasonable social dimension of shared accountability, so far lacking in other applications.

However, Planta lacks integrated diagnostic and subsequent advice found in other applications. These features provide an opportunity for the application to expand its domain from a data driven task allocation tool for houseplants to a more holistic system combining environmental modelling, species identification, and educational support.

2.3.4 Conclusion:

In summary, existing plant care applications demonstrate individual strengths such as large-scale data provision, real-time monitoring, and multi-dimensional data-driven scheduling. However, no single solution effectively integrates personalisation for both modelling and user engagement.

This project aims to bridge these gaps, combining adaptive care recommendations, environmental data, and ethically designed mechanisms for user engagement.

3 Tools and Methods

Table 2: Tools and Methods

Technology	Purpose	Pro	Contra
MySQL	Database design and implementation	• Free and open source • Widely Used • Prior experience	• Less flexible than other databases • Scaling can be more complex
PlantNet API	API for identifying plant species	• Eliminates for custom model. • Returns confidence scores.	• Dependence on third party application • Limited control over behaviour. • Requires internet connection
React	Frontend framework for dynamic interfaces	• Modular structure improves maintainability • Fast and efficient updates. • Well suited for mobile first design	• New tool with a learning curve for state management
Visual Studio Code	Integrated development environment	• Prior experience • Works well for web dev technologies • Extension support	• Performance can degrade with lots of extensions
HTML	Markup language	• Standard language for web structure • Prior experience • Integrates with react	• Limited functionality
PUG	Markup language	• Cleaner syntax than HTML • Supports template inheritance	• Requires a build step to run in browsers with react • Not compatible with react native

CSS	Presentation and design	• Widely supported • Well documented	• Can become difficult to manage at scale • Requires careful organisation to avoid conflict
Pytorch	Machine Learning	• Powerful for machine learning • Widely used • Well documented • Full control	• Training can be resource intensive • Requires a suitable dataset
Aseprite	Art/Design tool	• Simple to use • Excellent for pixel art	• Limited to only pixel art • Not free
Node	Backend runtime	• Good performance for I/O applications • Lots of libraries • Works with react frontend	• Not ideal for CPU intensive tasks • Asynchronous code can be hard to debug
Express	Backend Routing	• Simplifies backend development with Node • Well documented • Well supported	• Requires good organisation for scalability
AWS	Deployment	• Reliable cloud hosting • Supports scalable deployment • Industry standard	• Steep learning curve • Can be expensive if misconfigured • Complex dashboard
Docker	Running Containers	• Consistent environments • Simplifies deployment and setup • Modular system design	• Additional setup complexity

The table above demonstrates the strengths and weaknesses of potential tools to use throughout the development of the final artifact. This list is prone to expansion as potential issues/conflicts arise.

3.1 Rationale for the Technical Tools Selected

After careful review, the initial stack design will consist of:

Docker: To ensure consistent development and deployment.

Visual Studio Code: To develop the front end and backend code necessary for the website to function. The project space will be set up with GitHub for availability to up-to-date project files across various machines.

HTML: Used for markup language scripting due to wider compatibility than PUG templates.

CSS: Will be used for decoration and functionality. This is the most widely supported decorator for markup languages.

React: The frontend framework for dynamic interfaces, chosen due to its aptitude for mobile application development.

Node: An open-source runtime for web application development which is widely used, well documented and widely supported.

Express: This framework simplifies and organises the development of node.js. It will be implemented to reduce the necessary lines of code and simplify layout.

Aseprite: This application is one of the highest recommended pixel art tools for beginners. With plentiful documentation and support, it has a wide community of user led assistance.

Future Tools:

The above stack has been selected due to its synergy, but also for potential future development and deployment. Providing time permits, the project scope could expand to deployment and independence.

AWS: Amazon Web Service has 31% market share for cloud infrastructure as of Q1 2024 [17]. With its dominance over the industry, and free hosting features, AWS would be the ideal platform to initially deploy this project as a usable mobile web application. AWS also fully supports the stack outlined above.

PyTorch: PyTorch is an open-source deep learning library for python developers. To reduce dependence on a third-party API such as PlantNet, a model can be developed using PyTorch providing a sufficient dataset is available for training. Development of a custom model allows for full control, access, and offline capability.

3.2 Rationale for the Research Methods Used

3.2.1 Datasets and feature engineering:

Datasets were identified as a crucial method within this project. It is argued that there are 450,000 species of plants on Earth [18]. Obtaining information on care requirements for a vast quantity of species is an extensive task. Plenty of literature exists pertaining to ideal environments, care, and characteristics of plants, however there is no singular complete collection of information. The fragmented nature of such information imposes a challenge, where and how to find the data required? Fortunately, this is a problem others have also faced and attempted to solve. Kaggle is one of the largest global hosting platforms for data scientists [19], containing many varieties of datasets, including plant care datasets. One dataset was found to be most suited to the problem of plant care requirements. The dataset featured fields such as origin, Latin name, family, ideal light and plenty more. The dataset had a strong feature set however did not match the structure of the database schema for ingestion. To remedy this, feature engineering was executed, expanding on the dataset. Origin and climate data were used for semantic domain expansion using mapping. Origins and climate aspects were extracted and combined, then mapped to plants matching those fields.

This set of “ideals” was used comparatively with data pertaining to room environments creating recommendations on optimal location.

3.2.2 Care books:

The royal horticultural society is one of the largest organisations devoted to researching and growing plants across their five flagship gardens [20]. The RHS has an extensive list of publications in forms of blog posts, vlogs, tv programmes, books, and research papers. Throughout the development of the project one of their books was regularly consulted. “Royal Horticultural Society Practical House Plant Book” [21] was selected for its 175 different plant profiles with custom care guides, all structured in a reader friendly, comprehensible format. The data presented was relevant, followed a natural language structure and was translatable into a database structure. The literature also contained inspiration for the future development of a projects page, containing creative display projects, propagation projects, advice on shaping, and pruning. The book also contained useful guides on disease identification and prevention, a useful feature which could be later implemented.

The book was used primarily for identification of key data to design and build entities, as well as inspiration for future developments.

3.3 Project Management and Product Lifecycle

Project management methods were utilised, ensuring steps from ideology through to execution were implemented. Tasks were defined prior to execution and a development path devised. GitHub was used as the primary method for project planning and tracking. GitHub's Gantt chart and kanban board features proved to be most effective for tracking status of tasks, as well as defining order of execution and deadlines. The project was abstracted into components, then clustered into three areas: a project overview containing all tasks, a project artifact relating to the product development, and a project documentation section containing all the tasks and descriptions presented in this report.

Artefact tasks were defined as "User Stories", a way of presenting a problem from the perspective of the intended user, enforcing design focused on real user problems instead of checklists of requirements falling outside of scope [22]. These user stories were abstracted further and defined with implementation steps within the description. The most effective layout for monitoring deliverables was the use of kanban boards within GitHub. The kanban board used within this project featured four stations. When tasks were devised in the form of user stories, they were entered in the "To-do" station ready to be picked up, the second station was for tasks that were "In progress" whether it be the research or implementation stage. The third station was "In-review", containing any tasks which had reached an initial state of completeness, ready to undergo testing and review before reaching its terminating station "Done". The journey of a task through these stages provided a way to effectively pick up from where a task was last left. Once a task was completed, a new one from the "To-do" section was picked up.

The report utilised Gantt charts as the primary project management tool, with kanban boards as a secondary. Gantt charts are a way of managing time within a product's development lifecycle, assigning start dates and end dates to tasks to set durations and deadlines. Since the report is literature-based and follows a sequential, fixed and milestone-based structure, Gantt charts were selected as the primary project management tool. Items were defined by headings of sections within the report, further described by deliverables. Durations and deadlines for each item were assigned based on quantity and complexity of deliverables outlined within the description.

4 Legal, Ethical and Professional Issues

4.1 Legal and Regulatory

While designing an artifact with the intention of public use, it is crucial that for publication, the product is compliant with current and future relevant laws and regulations. To comply with mandatory legislation, legal acts need to be identified.

4.1.1 GDPR

GDPR is a legal framework introduced by the European Union to regulate the collection, processing, and storage of personal data. It primarily is in place to hold organisations accountable for transparent, lawful, and secure handling of individual's personal data. To remain compliant, organisations must follow the key principles of data minimisation, accuracy, and purpose limitation. This means that personal data being held is relevant, up to date, and limited to what is necessary for its intended use.

GDPR compliance was addressed through adherence to principles of data minimisation, secure storage, and user rights management, including access, correction, and deletion of personal data.

4.1.2 EU AI Act

The EU AI act was first proposed in April 2021, coming into force in August of 2024. The legislation was designed to ban AI systems posing unacceptable risk, introduce codes of practice, and rules on general-purpose AI[26]. The purpose of the regulation is to “promote the uptake of human centric and trustworthy artificial intelligence while ensuring a high level of protection of health, safety, fundamental rights as enshrined in the Charter of Fundamental Rights of the European Union,”[27]. To ensure compliance with the act, the intention of implementation of AI into this product should be defined, and its risk factor evaluated. The EU AI Act creates buckets for risk factors and implements standards for each of these buckets.

Under the framework of the EU AI Act, “Urban Botany” would be classified as a minimal-risk AI system. The project uses image data to determine plant species and provide a care guide to the user. The application does not make legal decisions, diagnose medical conditions, control physical systems, enforce rules on people, or perform biometric identification and therefore does not fall under high risk. The application at this stage does not have any intention of interacting directly with

humans, generate or manipulating content, or utilise emotion recognition or biometric categorisation, as such the project does not need to comply with the obligations of article 50.

Minimal risk AI systems have no mandatory obligations; however, the act encourages the implementation of voluntary practices such as transparency, human oversight, risk mitigation, and data governance.

4.1.3 Intellectual Property and API Licencing:

PlantNet:

For species identification, the PlantNet API has been utilized. PlantNet has its own “Terms of Use” which can be found on its website. To comply with the terms of service, the user must not commit to using any racist, pornographic, or slanderous words on the application.

- PlantNet must be acknowledged when using the API
- Only one account is permitted per person/organisation
- Commercial use requires a paid license

PlantNet states and warns that there is not a guarantee of accuracy, users are to use the API and its results at their own risk and that PlantNet are not liable for incorrect results, data loss, or financial or material damages.

Intellectual property of PlantNet is protected by intellectual property law, and as such content cannot be copied, reproduced, or redistributed without written permission. The API is permitted for use, but republishing of data or images is forbidden.

4.2 Ethics

4.2.1 Model Accuracy and Disclosure

As stated within the EU AI Act. As a minimal risk AI system, Urban Botany is not required to comply with the framework put into place for other risk categories. The EU AI Act does however advise that transparency, human oversight, risk mitigation, and data governance frameworks are implemented.

Transparency will be implemented through the provision of confidence scores. Human oversight will be implemented through a user validation prompt of whether the predicted plant matches the photo provided by the user.

4.2.2 Identification of Toxic Twins

Urban Botany will be designed with safeguards for plants with toxic counterparts to reduce risk of poisoning. Plants which are diagnosed as toxic to either humans or animals will be labelled as such within the database, displayed to the user appropriately.

Further steps for human oversight for diagnosing between toxic and non-toxic doppelgangers is another suitable approach which will be investigated during future works.

4.2.3 User Content Guidelines

To comply with PlantNet's terms of service, a user content guideline will be developed, with rules of appropriate, and inappropriate content.

To futureproof the product over interactive features such as friends, households, and other methods of sharing content. A terms of service should be drafted and published alongside the user content guidelines. Upon publication users would have to agree to Urban Botany's Content Guidelines Terms of Service to use the application.

User content guidelines would prohibit the publication or distribution of pornographic, hateful, or racist content. Distribution and publication of such content or violation of other terms will result in account suspension and/or deletion.

4.3 Risk register

Throughout the exploratory research process, reading relevant literature, exploring competitors, technologies, and legislation, a series of risks have been identified and categorised.

4.3.1 Legal Risk

The EU AI Act informs that there is no mandatory framework for minimal risk AI. The recommended framework however should be implemented to reduce any possible future risk.

The biggest legal risk within this project relates to compliance with GDPR. Unlike the EU AI Act, GDPR does not have different levels of compliance with differing requirements. Any organisation or individual collecting or storing personal data must comply. Throughout the design process, the product must be fully compliant to mitigate legal risk.

To protect against the risks outlined by GDPR, testing processes can be implemented post design to ensure integrity of design and security. Testing methods could include:

- Automated Compliance scanners
- GitHub Workflows
- Penetration testing such as white box and black box
- Ensure third party compliance
- Undertake internal audits

4.3.2 Reputational Risk

Many organisations are exposed to reputational damage at the expense of their customers/users. These organisations although not directly at fault, are held accountable for the actions of its customers/users. X (formerly twitter) has received high quantities of complaints after its AI Agent “Grok” was utilised by its userbase to generate pornographic images of other users [28], violating the company’s guidelines of “consensually produced and distributed adult content” [29].

To mitigate the risk of reputational damage, a user policy, and terms of service must be implemented. To further reduce reputational damage, filters can be implemented to detect harmful content automatically, as is used in many organisations. Web applications such as YouTube have implemented content scans which will check content meets its guidelines prior to publication, this order of operations dramatically reduces the risk of user’s being exposed to harmful content on its platform.

Reputational damage can be inflicted through lack of accessibility. To attempt to provide improved accessibility, testing can be implemented through manual testing, assistive technology such as NVDA, JAWS, and VoiceOver, and automated testing such as WAVE, and Accessibility Insight. These tests would be recorded and improvements implemented where necessary.

4.3.3 Product Failure

One major risk is the inability to provide deliverables outlined from the concept proposal. Species recognition provided the largest risk to the product. To mitigate this risk, a proof of concept was designed using the PlantNet API. The software was able to successfully identify a series of test images submitted to the API and return the results with confidence scores to the terminal. Throughout future development, accuracy and performance will be continually monitored. Post deployment, accuracy will be regularly tested and results stored in test tables, generation of confusion matrixes will be monitored and analysed.

4.3.4 Engagement

This risk imposes a range of potential issues. On one side of the spectrum lies lack of engagement, the risk of not using the product. On the other side of the spectrum is addiction. A midpoint needs to be identified and optimised for. To test engagement metrics, the following testing methods can be implemented:

- Monitoring click-through-rates
- A/B testing
- Average time on a page
- Logins
- Response to stimulus such as notifications

5 Requirements/Specifications

5.1 Personas:

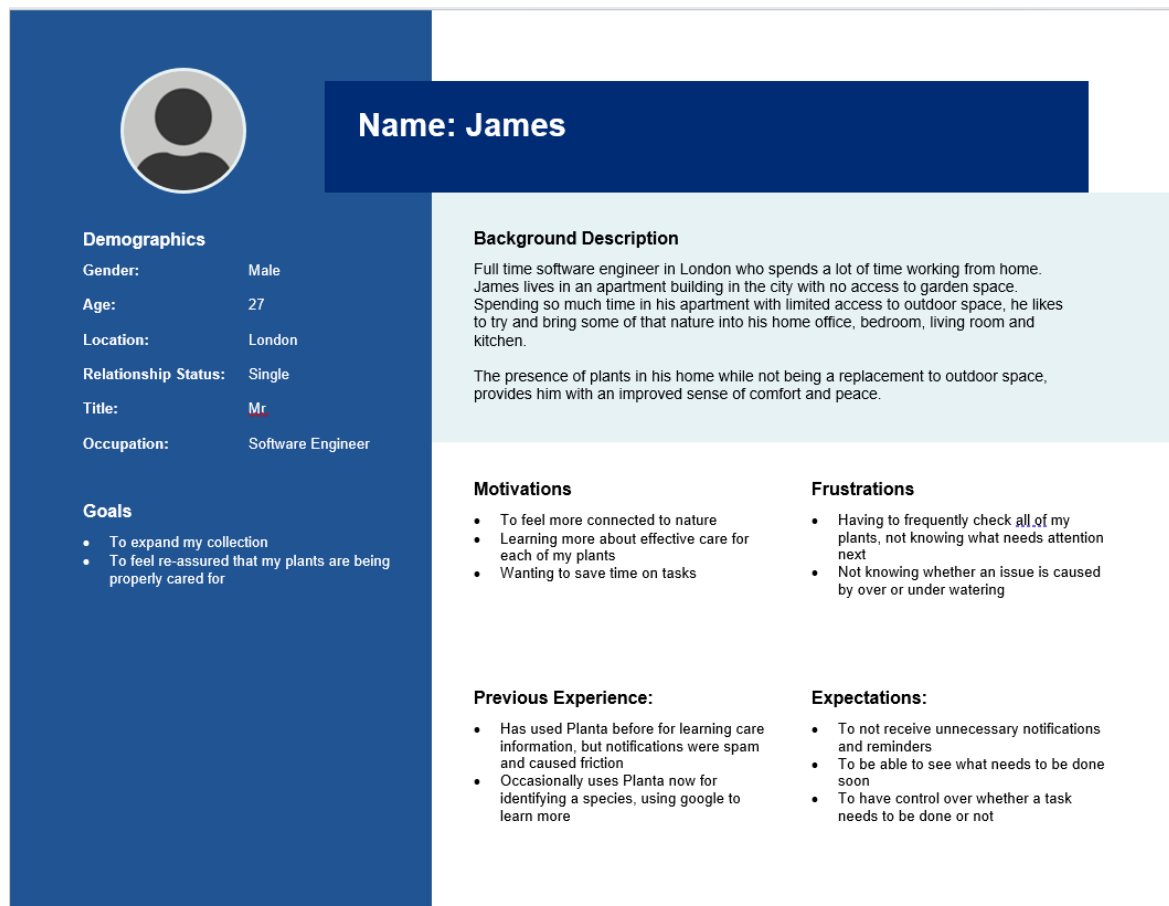


Figure 1: James's Persona

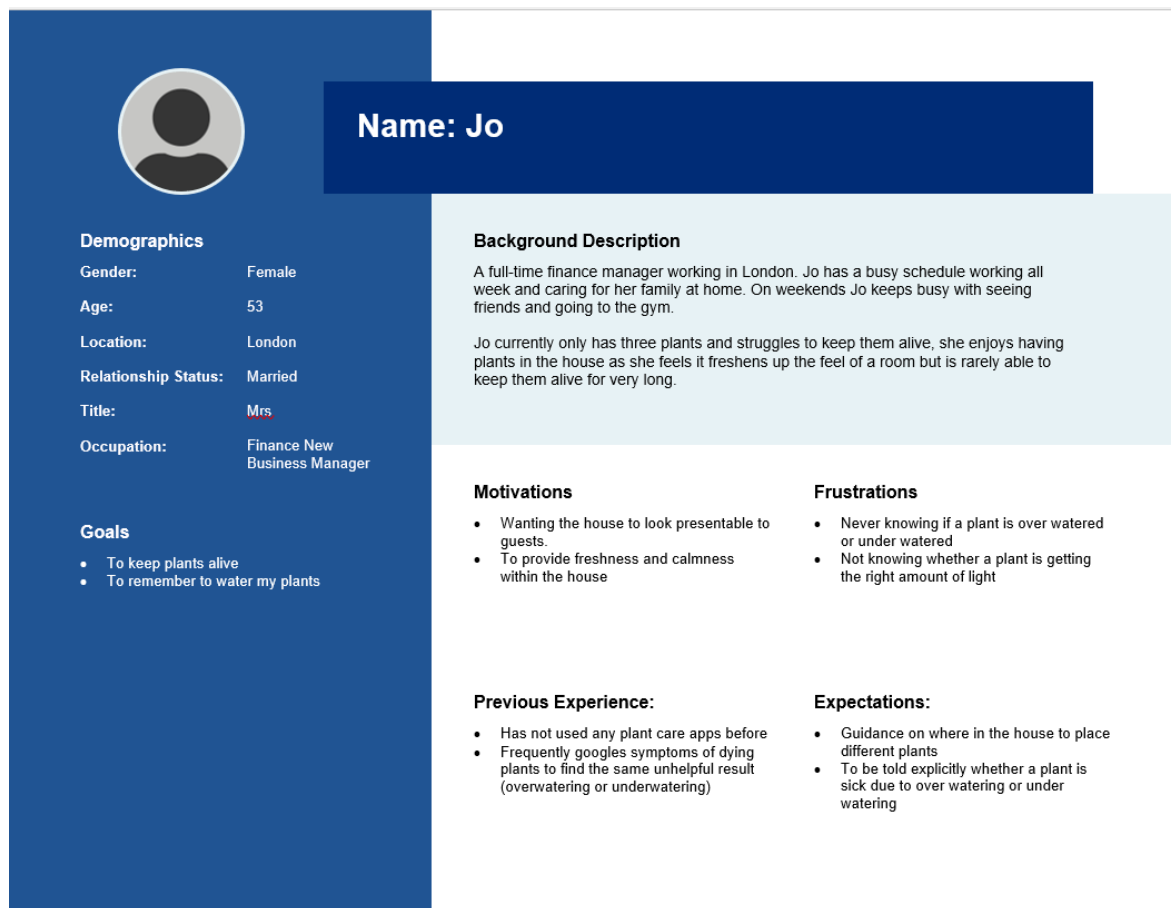


Figure 2: Jo's Persona

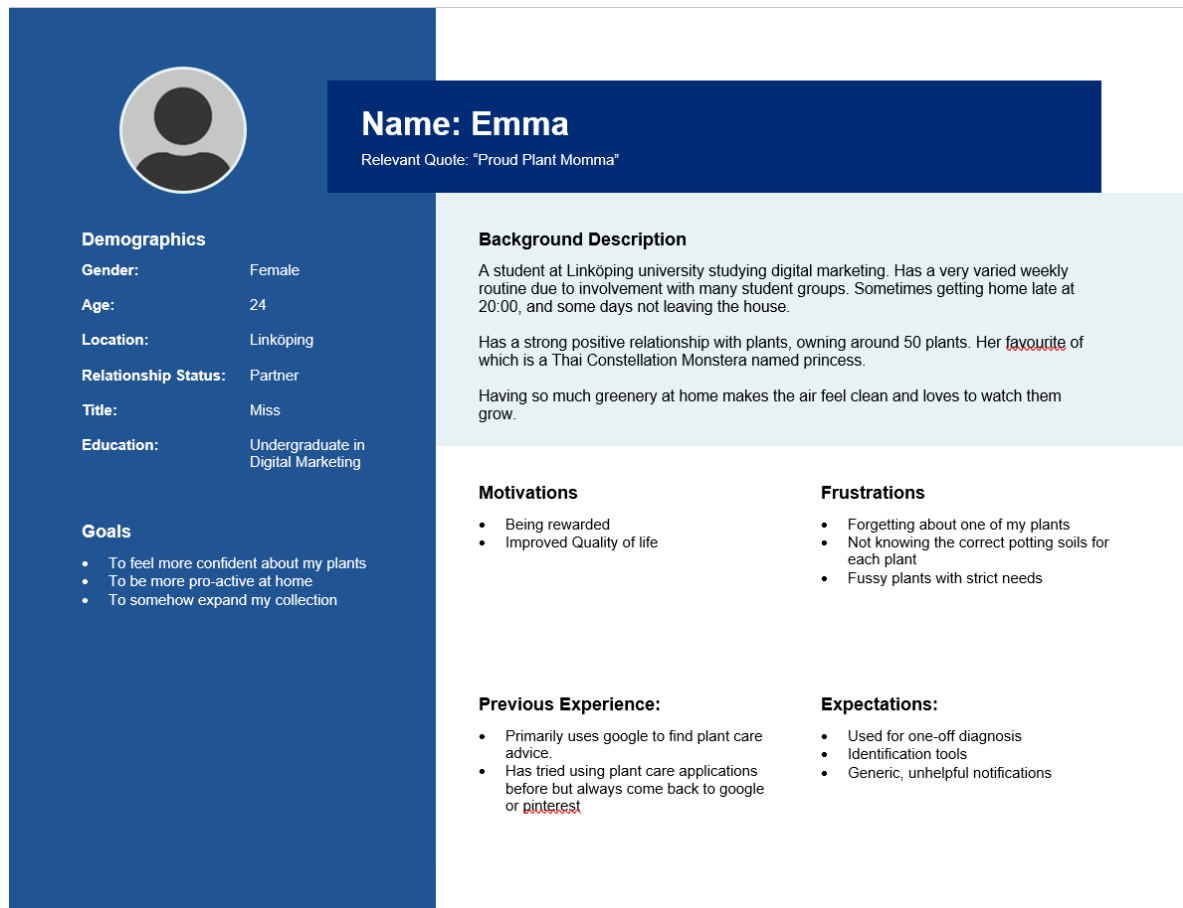


Figure 3: Emma's Persona

5.2 User Stories:

Table 3: User Story 1

User Story: As a person with a busy schedule, I want to be informed when to water my plants.
Implementation/Deliverables: • Create a dataset containing a list of plant species with their care parameters • Use these care parameters to create a “To-Do” list of tasks easily accessible and readable for the user.

Table 4: User Story 2

User Story: Living in a house, I have a lot of plants spread across different rooms. I want to be able to easily find the requirements of the plants inside of these rooms.
Implementation/Deliverables: • Organise plants by room, assign each user’s plant with a roomID so that they can be displayed on that room’s web page. • When a user loads that room, they can see the plants belonging to that room.

Table 5: User Story 3

User Story: Living at home, I have a variety of different rooms, I don't know what rooms are most suitable for each of my plants, how much light they should be receiving, whether they are warm enough or whether they are the right humidity.
Implementation/Deliverables: Store environmental data such as light, temperature, and humidity within rooms. Compare the room's environmental data against plant species specific requirements using methods such as Euclidian distance, regression or range-based constraint matching.

Table 6: User Story 4

User Story: As a novice plant enthusiast, I don't always know what a plant is or how to identify what species it is. I want a way to easily identify what a plant is.
Implementation/Deliverables: Build a convolutional neural network or use a pre-existing species recognition system such as the PlantNet API to identify plant species for the user.

*Table 7: User Story 5***User Story:**

As a keen plant enthusiast with lots of different plants, I want a way to be able to save and look at all of the information about my plants.

Implementation/Deliverables:

Create a relational database featuring a plants table and a user plant table. Plants relate to a species of plant and their relevant data, user plants relate to an instance of that plant species owned by a user.

5.3 Use case diagrams:

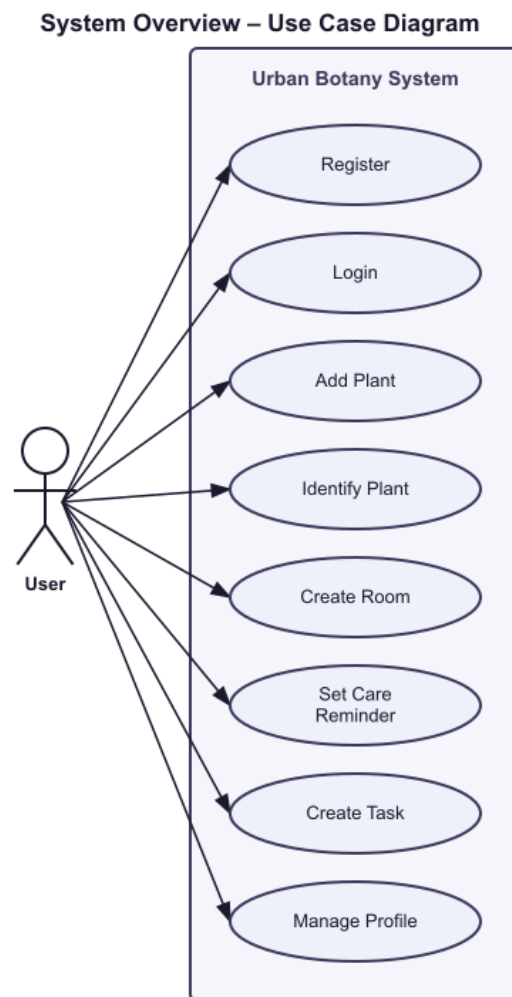


Figure 4: System Overview Use Case Diagram

The overview diagram in Figure 4 provides a high-level map of the app's primary use cases: registration/login, plant management, room creation, reminders, tasks, and profile management, grouped under "Urban Botany" with the user as the central actor. It's intended to quickly grasp core functionality and relationships between features. It is used to prioritise work and link diagrams to frontend routes and backend services. It's a planning tool for roadmaps and APIs.

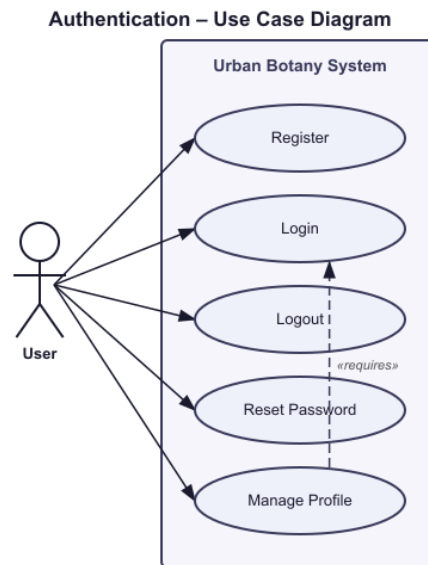
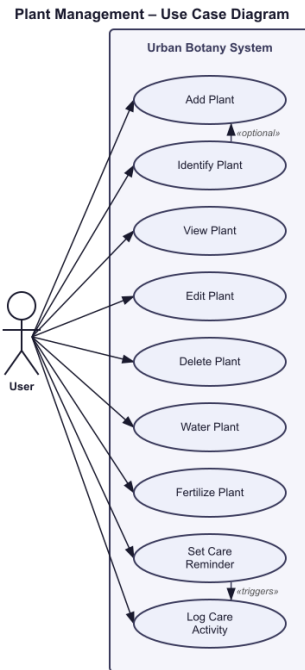


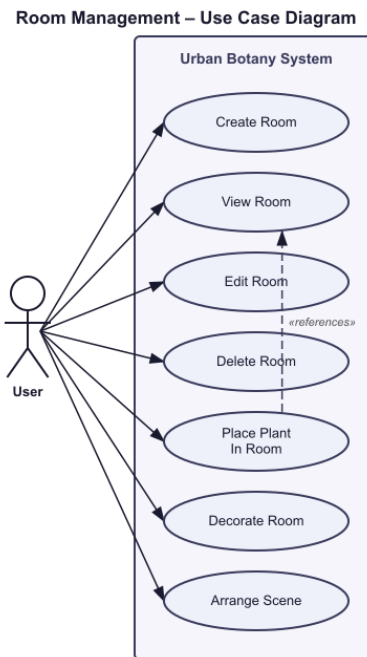
Figure 5: Authentication Use Case Diagram

The authentication diagram in Figure 5 shows the user's account lifecycle: registration, login, logout, password reset, and profile management. It emphasises authentication as the gateway for profile edits and protected actions. This view helps define required API endpoints, validation rules, and token/session handling. It's useful for scoping security requirements and test cases for conditions such as duplicate registration and expired sessions.



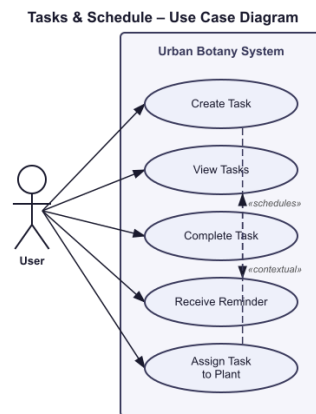
*Figure 6: Plant Management
Use Case Diagram*

The plant management diagram displayed in Figure 6 maps core CRUD and care workflows: adding and identifying plants, viewing and editing records, deleting entries, and logging care actions like watering or fertilizing. It shows optional identification leading to creation and how reminders trigger care logs. This clarifies data needs and relationships (species, photos, location, care history), UI flows for capture and identification, backend models, and automation for reminders. Use it to design API requests, backend routes, and database structure.



*Figure 7: Room Management Use
Case Diagram*

Figure 7 shows the room management diagram which captures creating, viewing, editing, and deleting virtual rooms plus placing plants, decorating, and arranging scenes. It models how users organise plants into location-based storage contexts and customise visual elements, and how plant placement references plant records. This informs entity and relationship design, endpoints for placement and asset management, and UI interactions. It helps define persistence rules for scenes as well as layer ordering priority for scene rendering. This guides both frontend component behaviour and backend storage logic.



*Figure 8: Tasks & Schedule
Use Case Diagram*

The tasks and schedule diagram in Figure 8 illustrates creating, viewing and completing tasks, receiving reminders, and assigning tasks to plants. It highlights scheduling and contextual links between tasks and plant records. This guides calendar UI design and recurring task logic. The diagram surfaces without highlighting potential backend and frontend divides such as backend task assignment with frontend user input for task completion.

5.4 Wireframes:

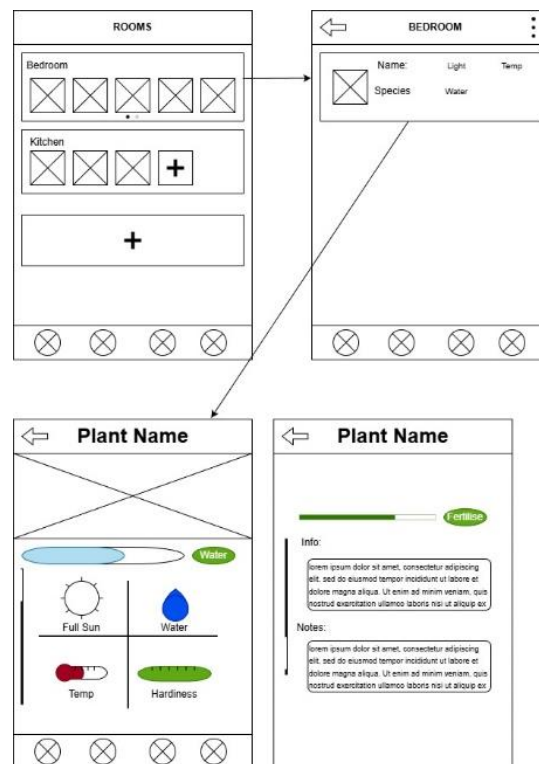


Figure 9: Rooms Wireframe

Figure 9 shows the rooms page, containing a user's personal rooms along with the journey they might take to navigate to a specific plant of theirs. The top left contains two instances of a room component, within these room card components are the user's plants displayed as images allocated to that room. Within these cards a plus button can be seen, allowing for a user to add a new plant to the selected room directly from the rooms overview page. The third component in the main body of the page is a large add button to create a new instance of a room. At the bottom of the page is the navigation bar to the web applications other root routes.

The top left view depicts a list of plants contained within a room. Once selected by the user, each plant card has an overview of basic requirement information. Upon selecting a plant, the user is directed to the plant's full profile as shown in the bottom two views.

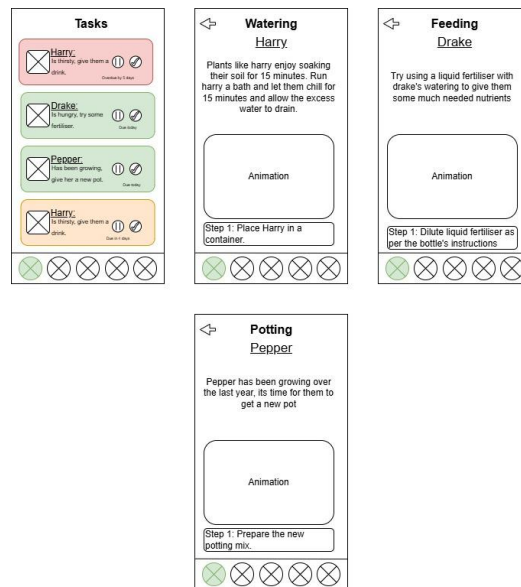


Figure 10: Tasks Page Wireframe

The task page depicted in Figure 10 demonstrates a journey the user might take to review care tasks to be undertaken. The top left depicts the tasks root route, an overview of all the tasks which are due, overdue, or upcoming as demonstrated by colour variations. A user can select the tasks card to open and review more information about the task as seen in the top middle, top right, and bottom view each with a different message and steps. The user can also interact with the tick button to confirm the task has been undertaken, removing it from the page until it needs to be completed again or select pause to temporarily hide the task for a few days without updating the task's history.

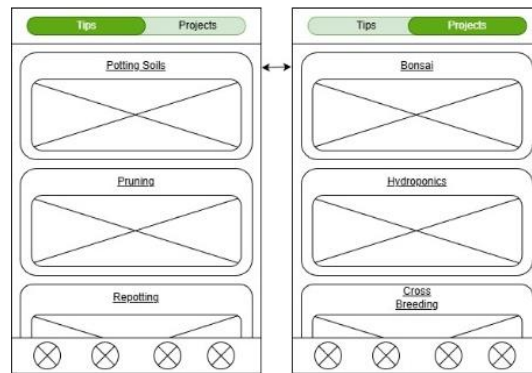


Figure 11: Projects Page Wireframe

The two views in Figure 11 show the projects and tips page functioning as a library of static pages containing tips for care or larger projects the user can undertake. The two categories are split into fields by a switch at the top of the page, illuminating the current view.

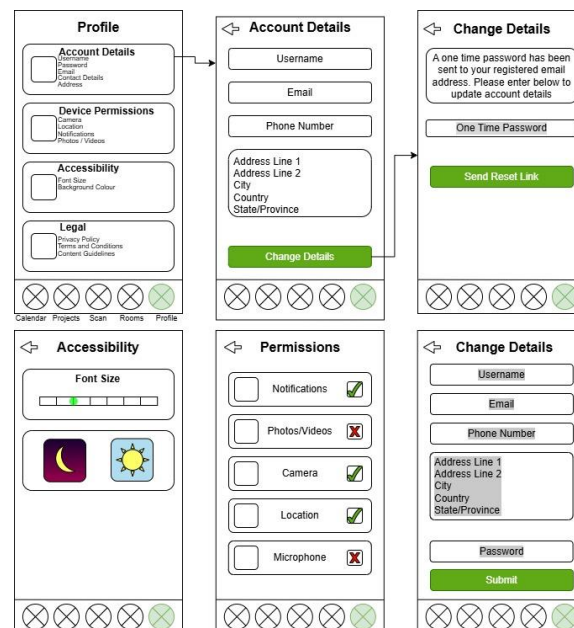


Figure 12: Profile Wireframes

The wireframes in Figure 12 depict pages relating to the profile routes. The top left view shows the root profile route, categorising all profile information into relevant sections such as account details, permissions, accessibility settings, and legal. Selecting a card will navigate to the corresponding page where the user can either review their account data, update their account data, change accessibility

settings, review permissions, or read a static legal page containing the web application's terms and conditions.

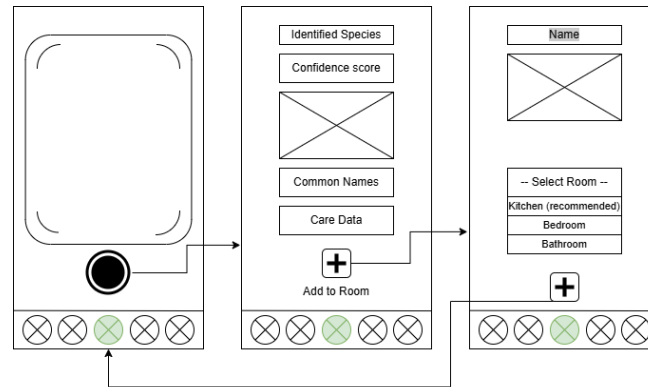


Figure 13: Scanner Wireframes

The scanner wireframes demonstrate how a user would identify and subsequently add a new plant to their account and room. Moving from left to right, the user will take a photo of the plant, displaying a page containing the latin name, the confidence score of the identification, the common names, care data and a button to add the plant to a room. Upon selecting the add to room button, the user is directed to the next page where they can allocate a personal name to the plant and assign it to one of their rooms. The backend handles logic for room recommendations by comparing room data to plant species care requirement data, providing a recommendation to the user.

6 Solution design

6.1 Database Design

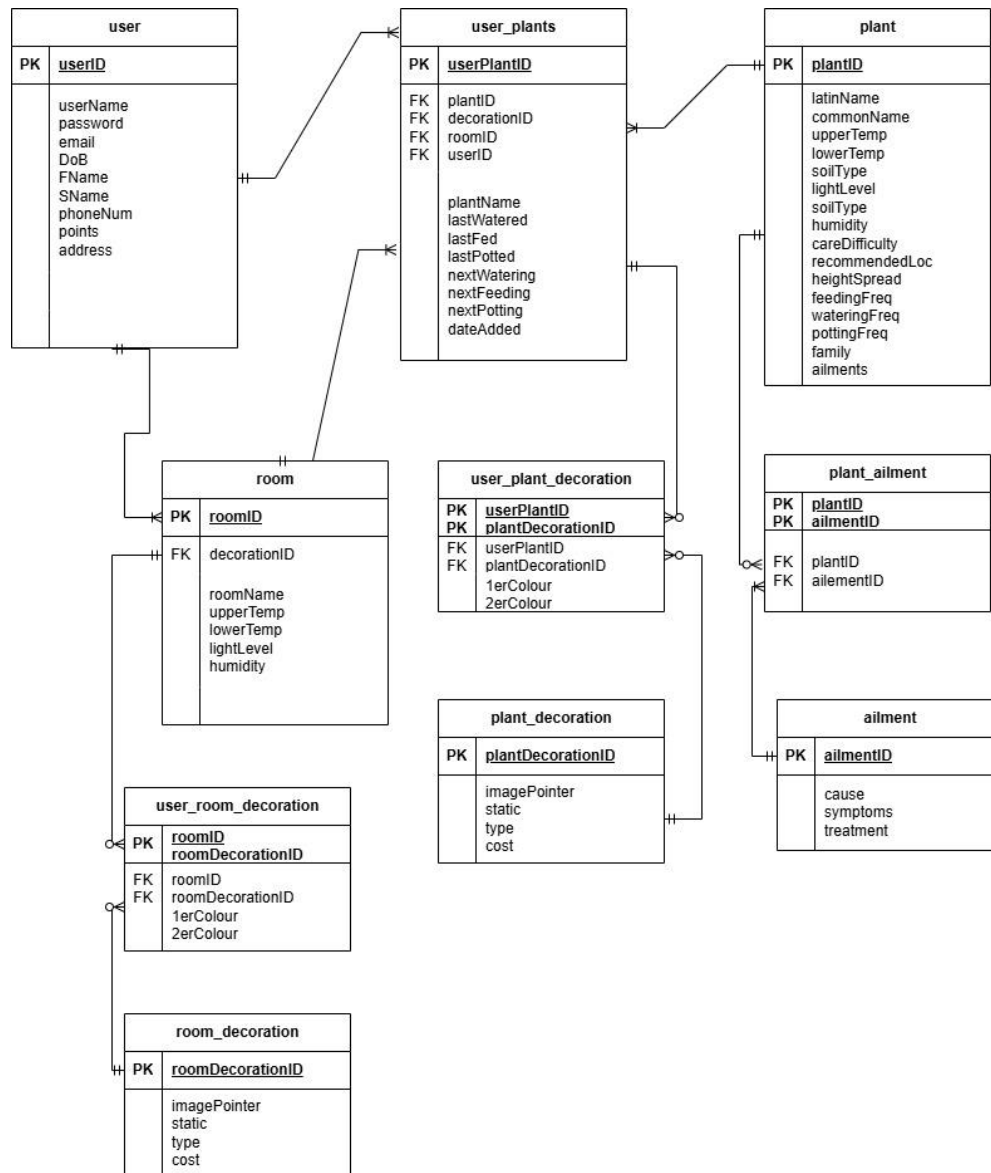


Figure 14: ERD Diagram for Urban Botany

Figure 14 shows the initial design for the ERD of the MySQL database of Urban Botany. The database features 10 tables, 6 of which represent core entities and 4 representing junction tables for instances of an entity.

Crow's feet notation has been implemented to demonstrate the relationships and dependencies between each table.

Entity tables are as follows: User, containing data relevant to an instance of a user; Plant, containing data relevant to a plant species; Room, containing environmental data about a room; Ailment, containing data about plant diseases/pests; decoration tables containing similar data for different expressions of a decoration type. The remaining tables are junction tables, containing an instance and belonging of an entity, with some additional fields important to that instance.

After some iterating, data fields were refined and expanded in some cases and naming conventions adjusted to prevent errors during backend implementation, delivering the design shown in Figure 15 below.

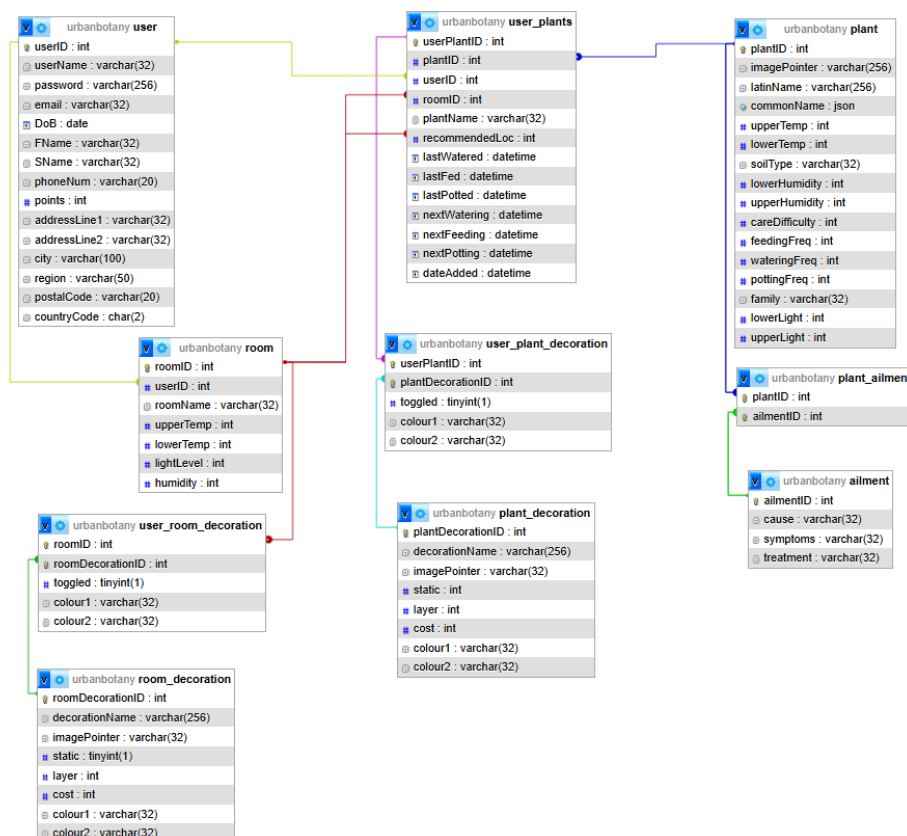


Figure 15: Final Database Schema

6.2 Sequence Diagrams:

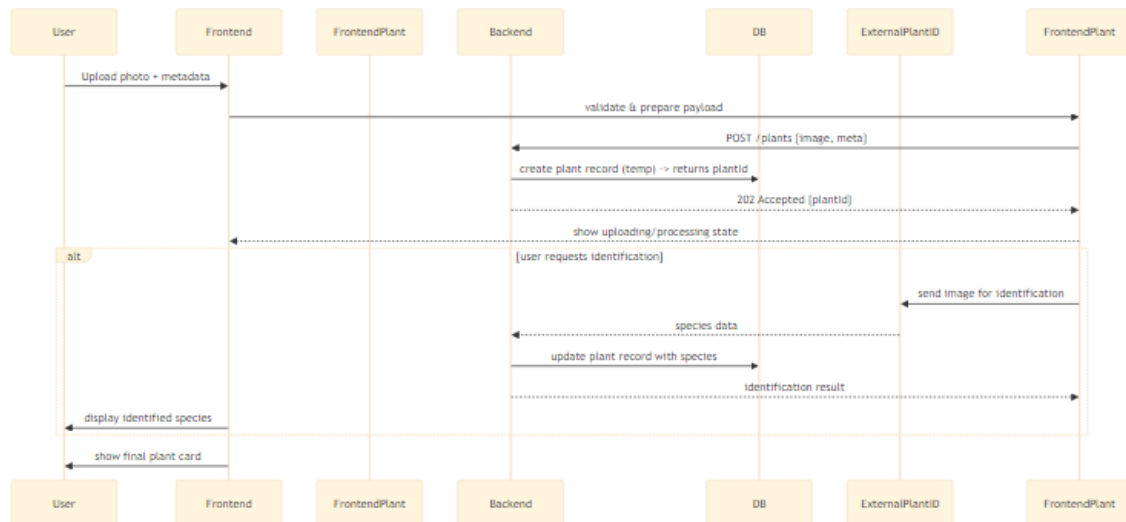


Figure 16: Plant Identification Sequence Diagram

Figure 16 explains how a user uploads a plant photo and metadata; the frontend validates and forwards the payload to the backend which creates a temporary plant record and returns a processing acknowledgement. If identification is requested, an external identification service is invoked, and its result is merged back into the plant record. The frontend updates the UI through the upload, processing, identification, and final card states.

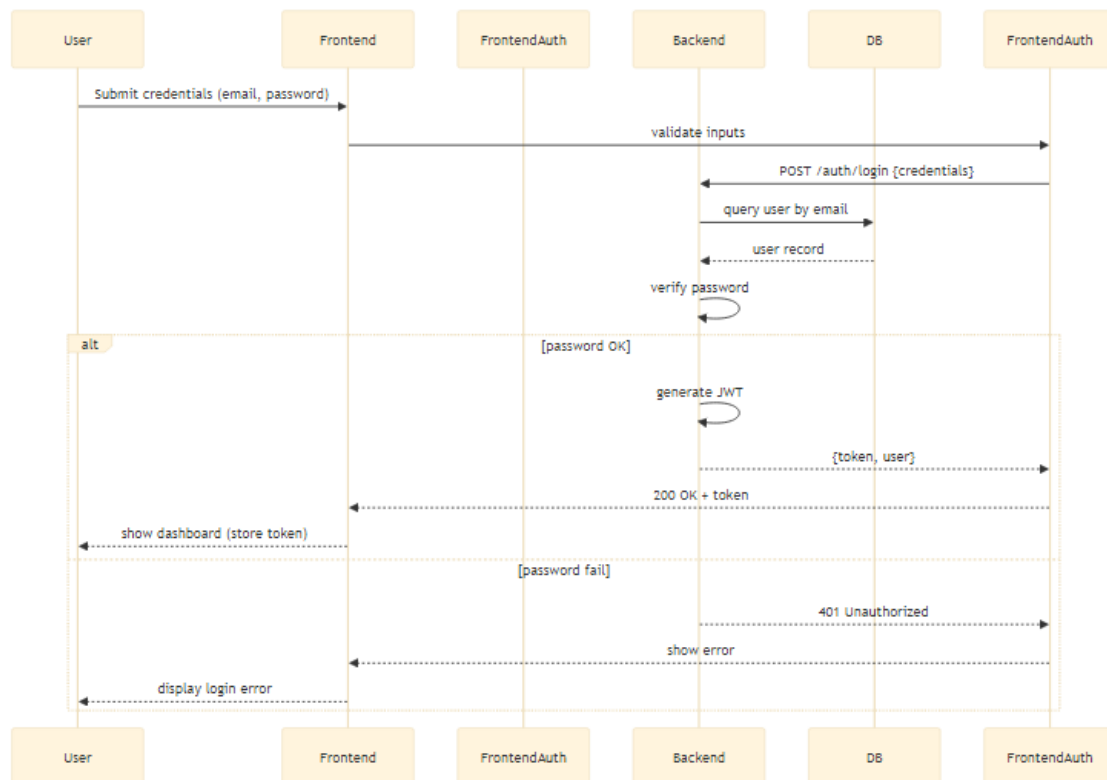


Figure 17: Authentication Sequence Diagram

Figure 17 shows the login flow of a user. The user submits credentials; the frontend then validates input before forwarding them to the backend which verifies credentials against the database. Upon success, a JWT is issued and returned, allowing the frontend to store the token and transition to the dashboard. On failure, the backend returns an error, and the frontend surfaces a clear message. Use this to validate auth endpoints, token handling, and error paths.

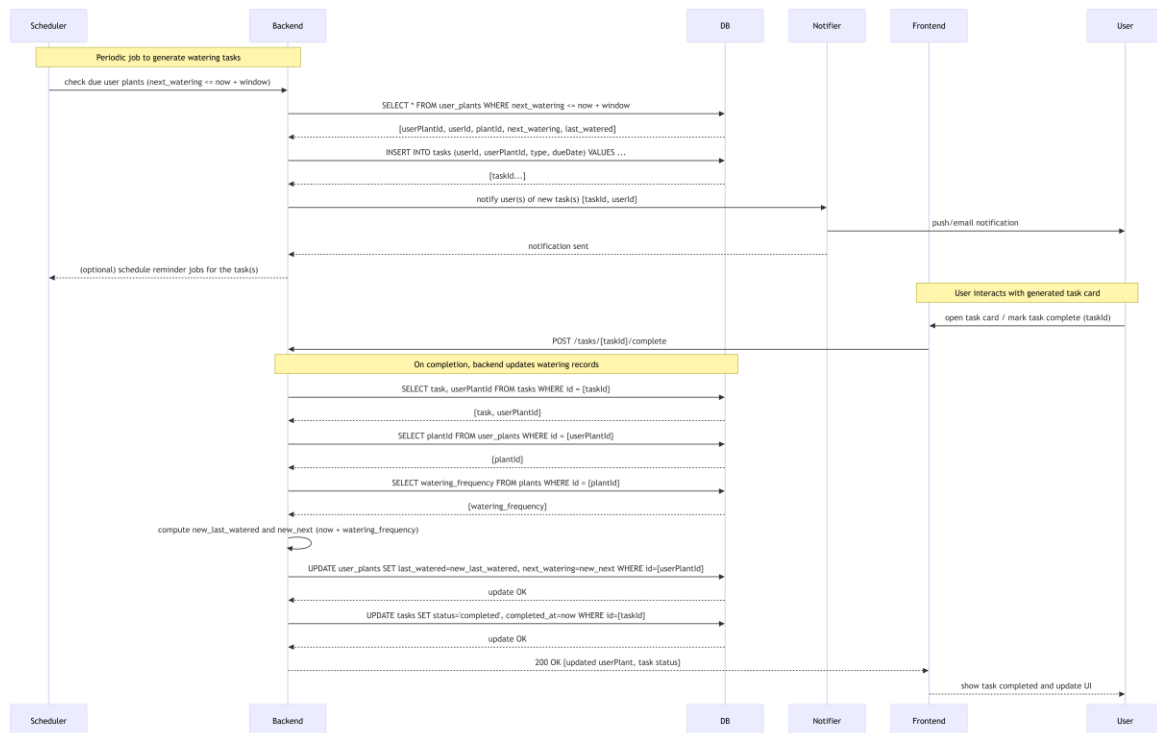


Figure 18: Task Generation Sequence Diagram

Figure 18 demonstrates how tasks such as watering, feeding, and potting are generated automatically. When a task is completed, the backend computes and updates 'last_watered' and 'next_watering' using the species' watering frequency from the plant table in the database.

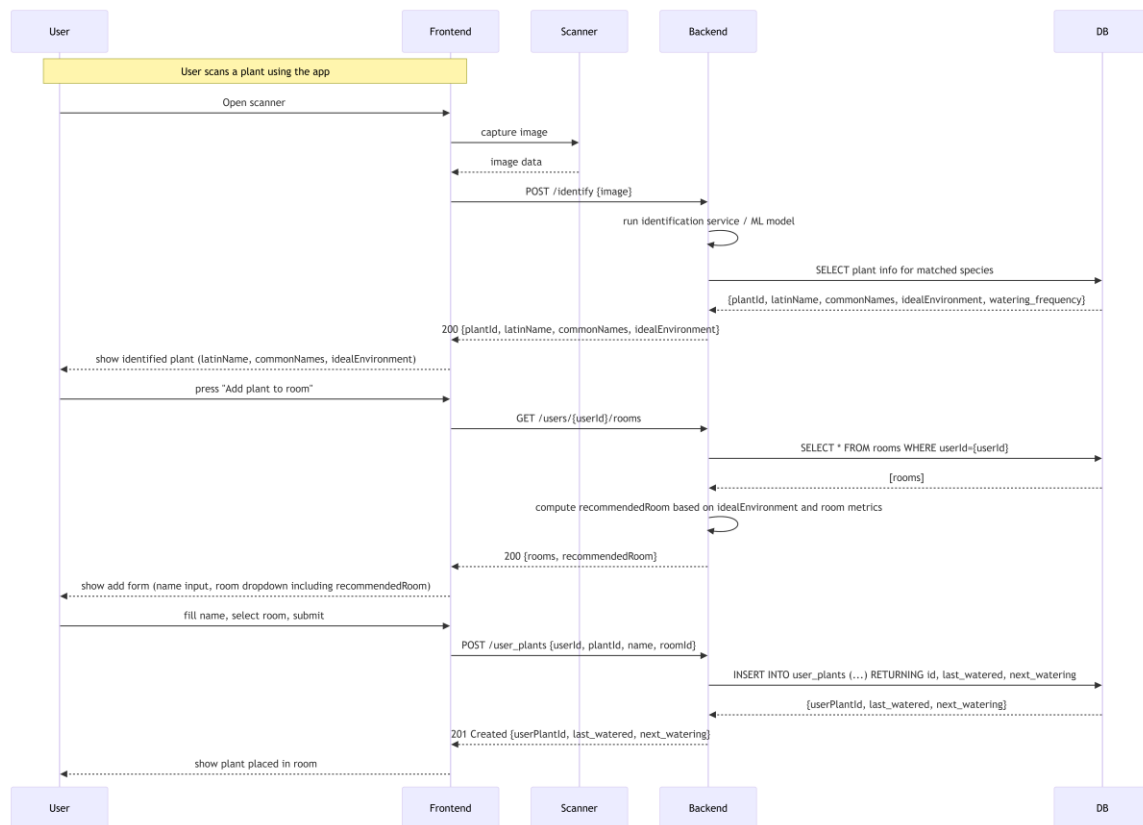


Figure 19: Plant Entity and Assignment Sequence Diagram

Figure 19 shows how to add a plant to an account, the user first takes a photo of the plant which is formatted appropriately before being sent for identification. The identified species is searched for within the plant database and species data is returned and shown to the user. The user can then add the plant to one of their rooms. A list of the user's rooms is retrieved along with the environmental data. The backend then runs a recommendation algorithm to determine the most suitable room. The user is shown a list of all rooms, with the recommended room flagged as "recommended". Once the user selects a room, a new "user_plant" is posted to the database.

7 Solution implementation

Transitioning from the planning stage to the implementation stage included a mediatory phase of sprint development. Sprints are a way of breaking the problem down into stages with deliverable components.

Sprint one was centred around setting up the environment and included creating the relational database in accordance with the schema outlined in previous sections. Separate docker files were created for the frontend, backend, mysql database, and PHP My Admin. Docker was implemented as a containerisation solution for efficient and repeatable deployment across systems. PHP My Admin was employed for database design through a user-friendly interface instead of pure SQL code. At the end of sprint one, the database structure was deployed inside of its container.

Sprint two focused on backend development. Routing was used to organise the backend structure as well as reduce conflict risk during development arising from route ordering. A layered architecture was selected for the project containing directories for routes, controllers, services, repositories, and entities. When a request enters the backend, it travels through these layers. The routes layer maps HTTP requests to the relevant controller. The controller requests an action to be performed using “req, res” parameters. The service then handles logic and validation before handing the request to the repository, the repository will send an SQL query to the database, which is mapped to an entity, travelling back up the path. This approach was excessive for a project of this size as it currently stands, but it incorporates security through design with layers for validation and authorisation. The architecture is also designed for scalability, allowing for improved efficiency in expansion without extreme refactoring. To test GET and POST requests, postman was installed locally, successfully showing results from GET requests such as retrieving plant data, and POST requests such as retrieving a session token upon login/signup.

Sprint three was heavily data oriented, taking a houseplant dataset and applying domain expansion to fit its original fields to those defined within the application’s SQL database. A Jupyter notebook was used to deploy feature engineering techniques for heuristic data. New desired fields included: humidity, care difficulty, watering frequency, feeding frequency, and soil types. Humidity was determined by mapping a plant’s known climate to average humidity ranges for those known climates, watering applied mapping from the datasets watering field, converting the string to an optimal moisture, minimal moisture and watering frequency to reach the optimum. Soil type ran

comparisons of a plant's climate, watering, and category for rule-based association with the most common soil types. The most complex additional feature was care difficulty. The method used was to assign a care score based on two identified categories of data dubbed "frequency data" and "range data". Range data primarily consists of environmental characteristics such as light, humidity, and temperature with upper and lower bounds. A score is created by taking the upper bound and lower bound of the entire dataset to define an overall range, and the upper limit and lower limit of a plant, the score is expressed as the percentage of the total range occupied by a plant's range. Frequency scores were determined by how much attention a plant needs, the function to generate this score. Frequency and range scores work in opposite ways, a higher range is more optimal, whereas a higher frequency is less optimal. The function takes the dataset's minimum and maximum values along with the plant's task frequency. It calculates and normalises the range, then uses this to form a penalty divider. The task frequency is divided by this value to produce a penalty, which is subtracted from the maximum score to give the final frequency score. Once these fields had been expanded, an insert SQL query was created and applied to each row of the data frame, uploading the new formatted data into the plant table of the urban botany database.

Sprint 4 focused on frontend development, personally the most challenging aspect of the artifact, it involved learning a new, commonly used JavaScript library called React. React allows for building dynamic and interactive frontend interfaces for users. React breaks down a page into components for improved render speed and re-usability. Learning new methods for combining html rendering with JavaScript functionality was initially challenging, but online resources such as YouTube aided in the learning process. Mobile-first design introduced challenges such as excessive white space on wider screens such as computers and laptops, making UI and UX design more challenging. The largest hurdle within this sprint was the rendering system which was devised. The application is designed with the intention of users being able to upgrade entities such as rooms and plants with decorations. A modular approach to upgrading rooms was devised by breaking a scene into different layers and rendered over the top of each other in priority order, this way users can swap out backgrounds, walls, windows, windowsills, weather animations, and curtains. A similar method was used for plants and plant pots with fewer layers and rules. The "user room decoration" table from the database is checked upon rendering, to determine whether a room does have any decorations, if a decoration exists, then the image pointer is returned and used during the layer building process, if no decoration exists for that layer, a default/fallback image is rendered in its place. The logic of this rendering method is reflected in the flowchart below.

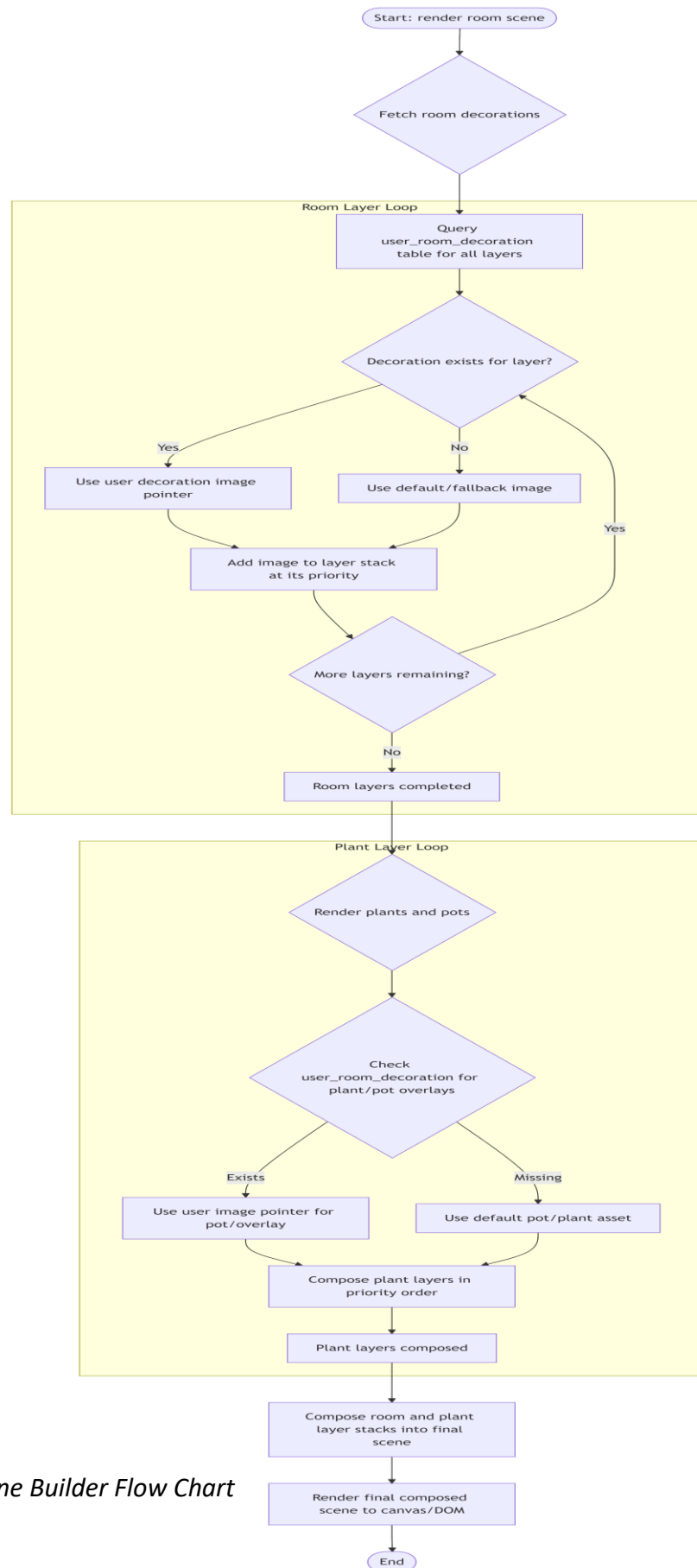


Figure 20: Scene Builder Flow Chart

8 Solution testing/ evaluation

Testing, evaluation, and development was implemented in an iterative flow. Following agile methodologies to continuously improve the quality of the product. The product was planned, designed, implemented, tested, and reviewed repetitively.

Think-aloud usability testing was deployed, providing users with a set of tasks to complete on the site, speaking out loud about what they were doing as they were doing it, along with any difficulties they were facing and thoughts they were having. Tasks assigned to users are as follows:

1. Sign up for a new account using the provided dummy user details.
2. Create a new room within your account
3. Add a plant to one of your rooms
4. Sign into the second account which has been provided to you.
5. Customise features of a room.
6. Check upcoming tasks and complete them.
7. Purchase a decoration for a room and for a plant and then decorate that item.

8.1.1 Sign up for a new account using the provided dummy user details:

Users were tasked with signing up for a new account.

8.1.1.1 Findings:

Users identified several experience issues regarding the registration interface. The form lacked visual styling, resulting in an unengaging and untrustworthy appearance. Additionally, the placement and similarity of the submit and sign up buttons caused confusion, as both used identical formatting. Excessive white space further contributed to a perception of incomplete design associated with lack of trust.

8.1.1.2 Action Taken:

The registration interface was redesigned to incorporate styling, improving clarity. A Landing page was introduced providing new users with background context and clarification of the purpose of the application. Primary, secondary, and tertiary colours were identified and allocated from the landing

page's concept artwork, applied consistently across the site. Button differentiation was improved through clearer separation.

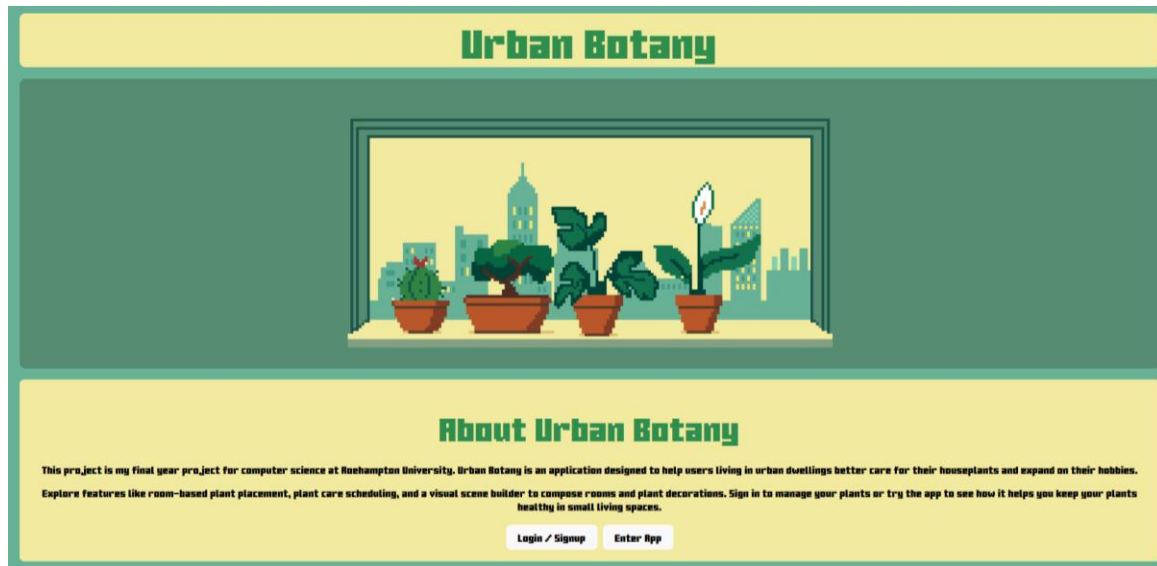


Figure 21: Landing Page

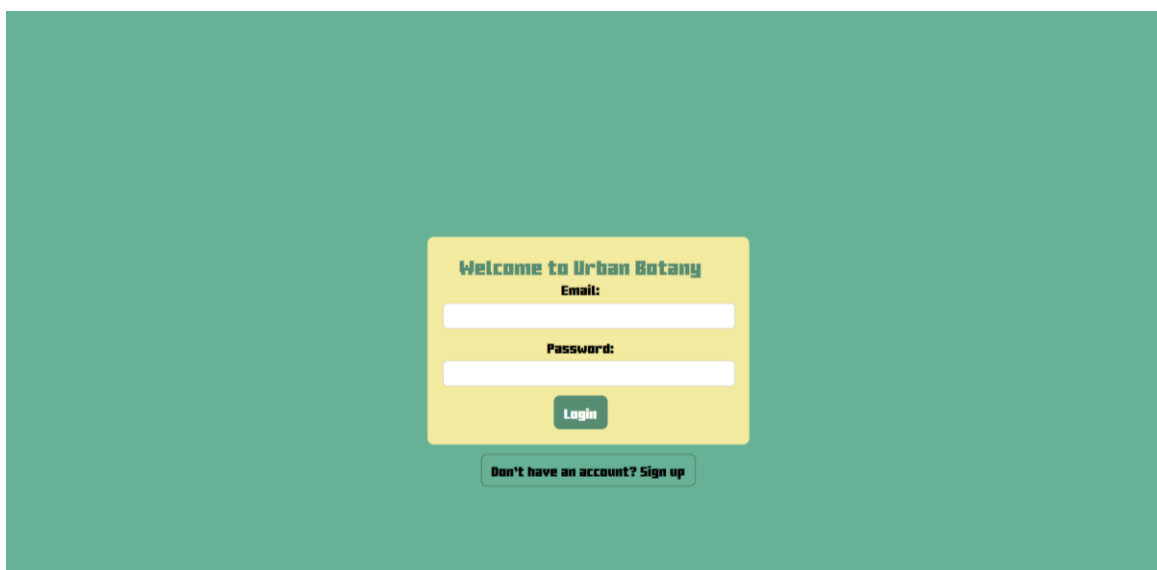


Figure 22: Login Page

8.1.2 Room Creation:

Users were tasked with creating a new room, within their account.

8.1.2.1 Findings:

Testing revealed issues with navigation clarity and interface comprehension through signposting. The action for adding a room was not clearly labelled, reducing discoverability. Additionally, upon discovery of the empty room page, the purpose of its existence was unclear, indicating insufficient signposting. With the application designed based on a fixed mobile resolution, when accessed on different devices, navigation bar width was not adaptive, limiting its usability, and the overall interface suffered from excessive white space, reducing visual engagement.

8.1.2.2 Action Taken:

The interface was redesigned to improve clarity and visual structure. Clear labelling and signposting were introduced to guide user actions and further discoverability. Adaptive constraints were applied to navigation bar components to improve consistency. Additionally, the application's unique colour scheme was applied with semantic justification, to indicate distinction between buttons, forms, and layouts.

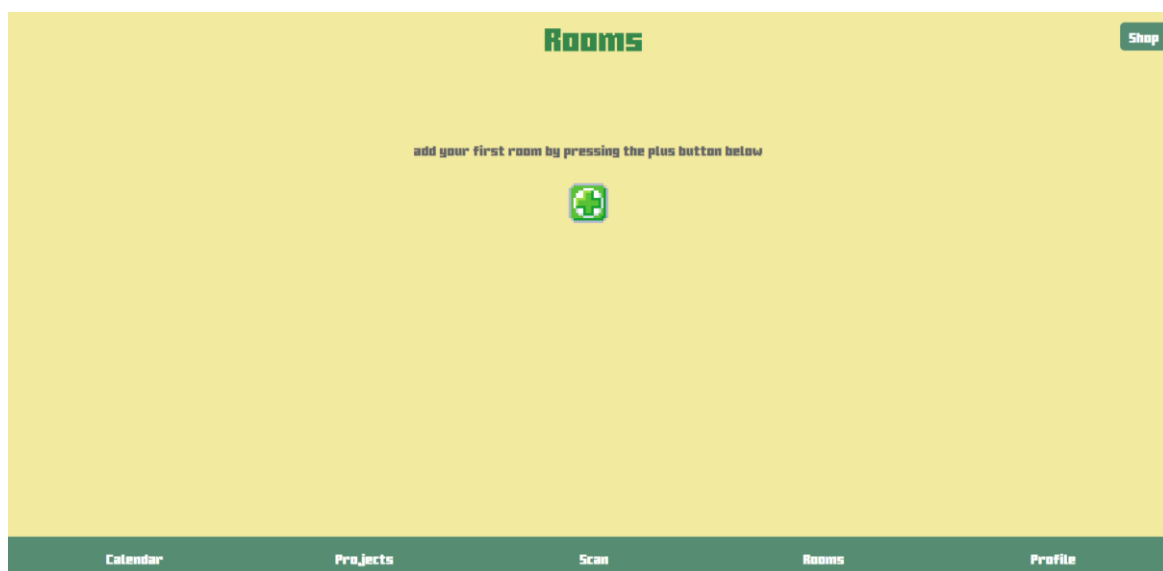


Figure 23: Empty Rooms Page

Rooms

[Shop](#)

Add a Room

Set environmental targets so plants can be matched to the best space.

Room Name e.g. Sunroom	Upper Temperature (C) e.g. 25
Lower Temperature (C) e.g. 18	Light Level 1 to 8
Humidity (C) e.g. 65	

Add Room?

☒ ☐

[Calendar](#)[Projects](#)[Scan](#)[Rooms](#)[Profile](#)

Figure 24: Create Room Form

8.1.3 Add a plant to one of your rooms:

8.1.3.1 Findings:

User experienced difficulty locating the plant scanning feature, initially attempting to add plants directly from the room page. This indicated a misplacement of discoverability features arising from improper understanding of the user journey. Additionally, the visual design of the camera clashed with the rest of the application.

8.1.3.2 Action Taken:

To improve ease of discoverability, a dedicated “Scan Plant” button was integrated directly with the room interface, aligning design features with user expectations. The user interface was also aligned with other application components for improved consistency, improving user experience.

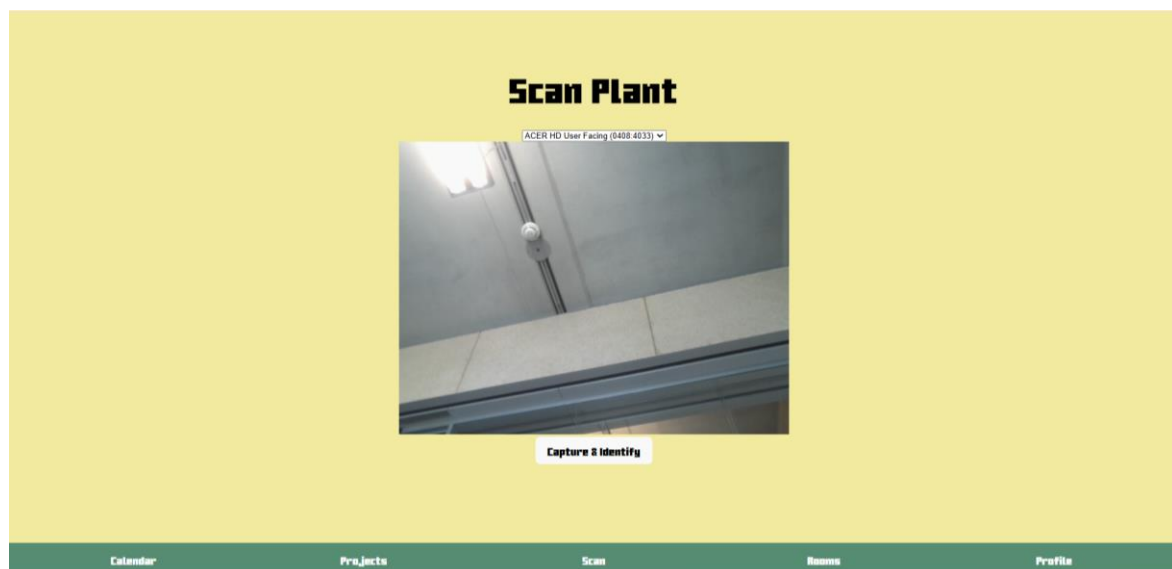


Figure 25: Original Scan Page

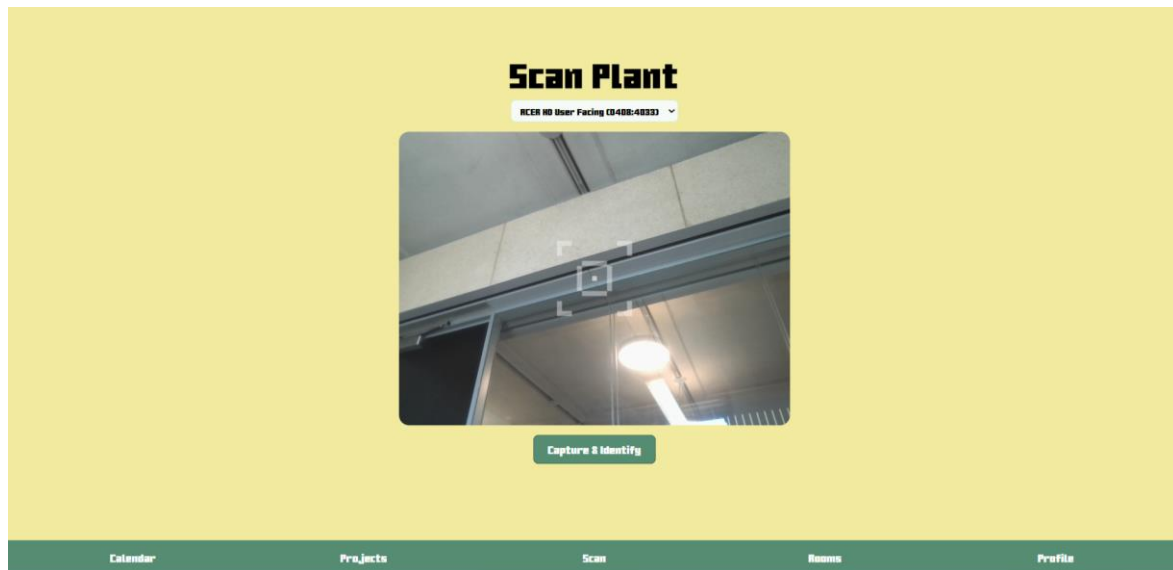


Figure 26: Improved Scan Page

8.1.4 Logging into an existing account:

Users were asked to log into a pre-configured account.

8.1.4.1 Findings:

A critical usability issue was identified in the absence of a visible logout function, requiring users to manually navigate away from the interface to sign out. This indicates the absence of a crucial account management feature. Additionally, one user initially attempted to log in using a username rather than an email address, suggesting a mismatch between user expectations and system input requirements.

8.1.4.2 Action Taken:

A dedicated logout button was introduced within the profile interface to provide clear and accessible account control. The login system was also adapted to accept both username and email as valid identifiers.

8.1.5 Room Customisation:

Users were asked to customise a room using available decoration features.

8.1.5.1 Findings:

Users responded positively to the real-time room preview, which provided immediate visual feedback and supported intuitive interaction. However, navigation issues were identified due to the lack of a back button for backwards traversal. Additionally, a user attempted to re-order plant positions inside of the interface, indicating the user expectation of a non-existent feature.

8.1.5.2 Action Taken:

Navigation was improved by introducing a back button on non-root pages for backward traversal. The expectation of plant arrangement was identified as a potential area for future development, highlighting an opportunity for feature and customisation expansion.

8.1.6 Task Management:

Users were asked to locate and complete scheduled tasks within the application.

8.1.6.1 Findings:

Testing revealed issues related to signposting and system feedback. Users stated the “Calendar” label did not clearly reflect to purpose of the interface, leading to difficulty in location tasks. Additionally, users were uncertain about the purpose of the pause function and the outcome of completing tasks, indicating insufficient visibility of feature outcomes.

8.1.6.2 Actions Taken:

To address visibility of feature actions, feedback mechanisms were introduced to clarify the results of user actions such as visual indicators being added to communicate reward assignment with an incrementing point wallet following task completion, and temporary notifications being implemented to better explain effects of pausing tasks.



Figure 27: Original Calendar/Tasks Page

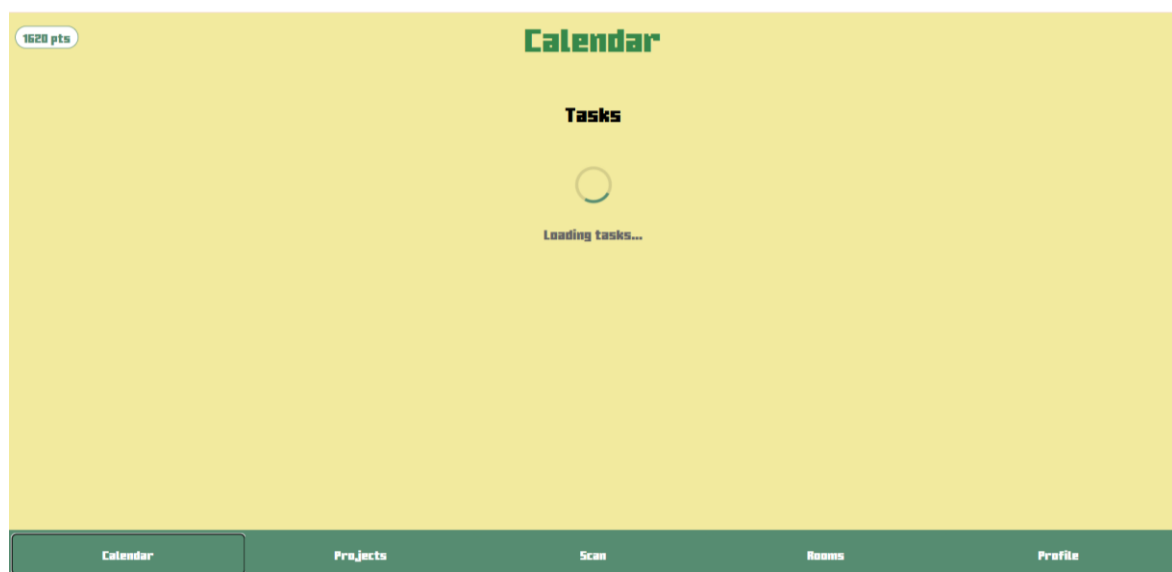


Figure 28: Tasks Page with State, Loading and Points

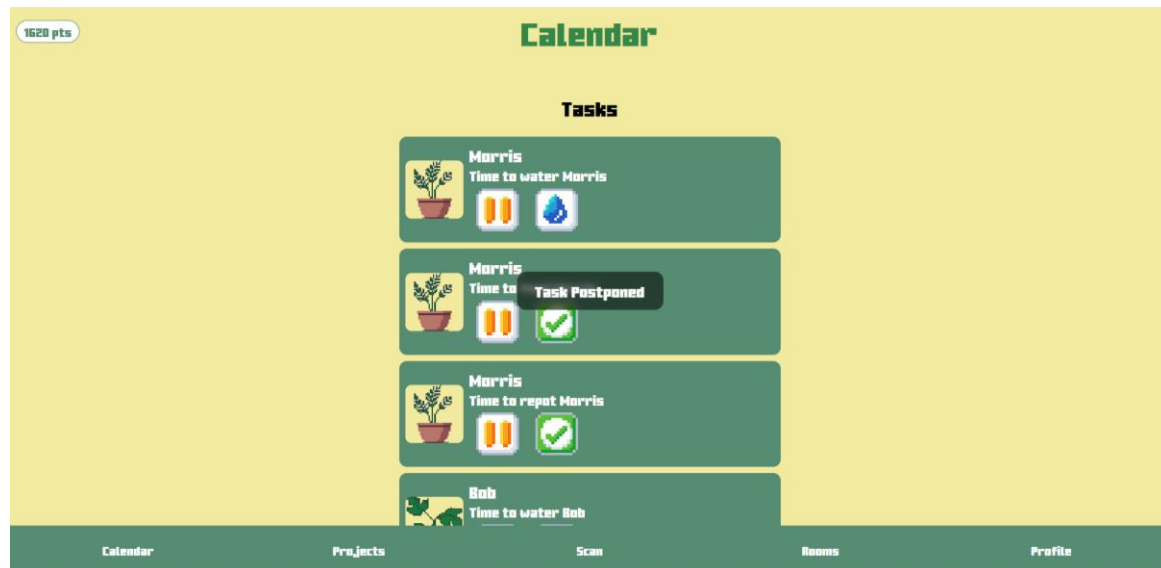


Figure 29: Task Page with Postponed Pop-Up

8.1.7 Purchase and apply decorations:

Users were tasked with purchasing a decoration and applying it to a room and/or plant.

8.1.7.1 Findings:

Several usability issues were identified relating to discoverability, interface clarity, and system feedback. Users experienced difficulty locating the store as the entry point had been isolated to root Rooms interface. Within the store, the filtering interface was not immediately intuitive, reducing immediate clarity of the purpose of the store, hiding items behind filtering layers. Additionally, delayed response following purchases created uncertainty regarding transaction completion. Finally, users found it difficult to track which item they were purchasing for.

8.1.7.2 Action Taken:

Store accessibility was expanded to better align with user workflows, improving discoverability. Interface clarity through more immediate confirmation of transactions, and improved product labelling. The item assignment process was identified as a design constraint; therefore, improvements were proposed to better communicate possession relationships during transactions while maintaining ownership rules for intended progression mechanics.

9 Discussion

This project is grounded in the premise that positive habits can be reinforced through repetition, supported by gamification mechanisms designed to sustain engagement. While such mechanisms are often associated with addictive tendencies, their application within an ethical framework can provide constructive self-improvement. This approach is reflected in established platforms such as Duolingo [30], Strava [31], and Habitica [32], which apply gamification methods across domains including education, health, and personal development. These systems demonstrate the ability of frequent and intermittent reward-based interaction models to enforce user engagement and subsequent positive habitual behaviours.

The effectiveness of gamification in supporting behaviour change has been widely studied, with evidence highlighting both its benefits in engagement and its limitations in sustaining long-term adherence [33]-[35]. Reward based mechanisms such as points, unlockable items, and streaks have been shown to improve engagement and retention, whereas penalty-based mechanisms can negatively impact user retention by introducing friction, discouragement, and loss.

User engagement and response varies when a social aspect is introduced within platforms. Strava introduces a competitive component which motivates some users while discouraging others [31]. This highlights the importance of adaptable engagement strategies, varying gamification mechanisms to better fit individual user motivations rather than a collective motivational model.

Initial user feedback indicates that Urban Botany has potential to support positive habit formation through its integration of ethically driven gamification methods. However, due to the limited evaluation period, long-term data such as retention, interaction frequency, and session duration was not available. As a result, long-term effectiveness of these mechanisms remains inconclusive.

10 Conclusion

10.1 Summary

This project aimed to address the challenges found in maintaining plant health in urban environments through the development of a user-centred application supporting plant care through the implementation of placement recommendations and care task management, backed by engagement mechanisms to improve consistency and habit formation. The application was developed with mobile-first design for improved accessibility and portability.

Evaluation through user testing demonstrated a demand and subsequent market does exist for such a product, and initial feedback suggests that gamification mechanisms improve upon experience and retention. However, limitations were identified in capture of long-term engagement data to consolidate some hypotheses.

Overall, the project has demonstrated that the combination of a problem area, relevant data points, and gamification principles can positively support behavioural change.

10.2 Recommendations for further work

Through the development of Urban Botany as an artefact, several areas for improvement and scope expansion were identified. Firstly, Willow provided enlightenment into the possibilities of IoT integration, creating custom devices which can provide backend services with improved data for modelling, and eliminating the risk of user-error upon entering environmental data.

Secondly, the application would benefit from the integration of longitudinal usage data, such as retention rates, interaction frequency, and session time. This would improve feature refinement to ensure mechanisms are functioning ethically, forming habits without building reliance and addiction.

Thirdly, the implementation of shared households would improve true-to-life functionality where shared households hold shared responsibilities. Users can have sole responsibility for plants under their possession, and shared responsibility for those in communal spaces.

Additionally, the pixel artwork within the application is hand-drawn, with so many species of plants, the task of curating artwork for each is laborious and time consuming. The creation of a automation

tool to generate on-the fly pixel artwork reduces time and can allow for further customisation with artwork better fitting user's plants.

Finally, the application would benefit from eliminating reliance on APIs for species identification. The current API would be used to label user submitted plant images, curating a new houseplant dataset which would be used to build a convolutional neural network for species identification.

10.3 Reflection

This project allowed me to develop a deeper understanding of web application development. Frontend development with React was a new challenge I undertook, creating dynamic fast responding web application interfaces. Additionally, the integration of APIs was applied and within the project. The implementation of a wide variety of tools and methods highlighted to me the importance of modular design and iterative development when managing a project with increasing scope and complexity.

Improved understanding of the development lifecycle for a full-stack application highlighted the necessity of early identification of core components and requirements for functionality. Aligning backend endpoints with frontend user interactions. One example of this would be the late-stage database alteration to allow for state with decorations, toggling decorations on/off. Such issues instilled the necessity of early user testing and feedback with prototypes and sketches.

The scope of the project unfortunately led to trade-offs between feature completeness and heuristic implementation. In hindsight, earlier prioritisation of areas such as AI implementation and long-term evaluation would have allowed for improved refinement, and increased data points for evaluation.

Going forward, I would place greater weight on early stages prototyping and user feedback, constraining scope and identifying design issues to be fixed earlier. As well as implementing an improved modular design structure so that components can be tested separately without reliance of other components.

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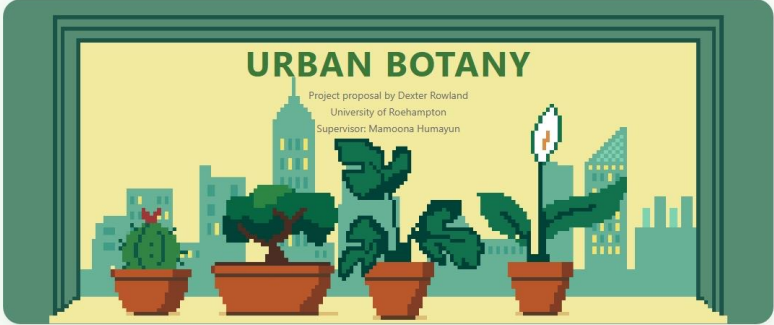
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12 Appendices

Appendix A: Project Proposal Poster

This appendix contains the original project proposal poster submitted at the start of this project. The poster outlines the original problem statement, proposed solution, intended methodology, application of AI, and some potential literature, alongside a potential timeline of implementation (created before deadlines were issued).



URBAN BOTANY
Project proposal by Dexter Rowland
University of Roehampton
Supervisor: Mamoon Humayun

Introduction
Urban environments limit access to outdoor green spaces; many city residents keep indoor plants but struggle with consistent care. **Urban Botany** provides a mobile-first solution combining care reminders, educational content, community sharing and light AI to help users grow confidence and keep plants healthy.

Problem Statement
Urban areas are constantly expanding, with more young people becoming isolated from nature. Considering the innate attraction that humans have to nature, it is no surprise that these urban dwellers are trying to draw nature into their homes through the care of house plants. However among the busy lives of these young people comes the struggles of knowing how and when to care for these plants, or how they can expand upon their hobby.

Aim & Objectives
Aim: To design and implement a mobile app that empowers urban residents to care for plants reliably, learn about plant science, and share practical foraging and recipe ideas.

- Investigate urban plant-care habits via surveys/interviews.
- Design intuitive mobile UX for plant scheduling and tracking.
- Integrate lightweight AI for plant identification and personalised tips.
- Test usability and measure engagement with a pilot group.

AI Component
Scope: Use a pretrained CNN (TensorFlow Lite / MobileNet transfer learning) to identify plant species from user images. The model will provide species name, basic care parameters, and confidence score. Results will be paired with database-driven care schedules and community-verified notes.

Implementation Plan

- Prototype with PlantNet / TensorFlow samples.
- Prune model for mobile inference (TFLite).
- Combine image results with rule-based care recommender.

Ethics
User images stored locally by default; optional cloud upload is opt-in. Data handled under GDPR-compliant policies; anonymised usage stats only.

Key Literature

- Wilson, E.O. — *Biophilia* (psychology)
- Krug, F. — *Don't Make Me Think* (usability)
- Sibley, E. — *Urban Botanics* (plant care)
- Russell & Norvig — *Artificial Intelligence: A Modern Approach*
- Lloyd, I. — *Information Technology Law* (GDPR and ethics)

Project Plan (Milestones)

Phase	Timeline	Tasks
Research & Definition	15 Oct – 10 Nov	Literature review, user surveys, ethics
Design & Prototyping	Nov – Jan	Wireframes, Figma prototypes, usability feedback
Implementation	Feb – Mar	Build app MVP, integrate AI module
Testing & Evaluation	Apr – May	User testing, metrics, refinement
Write-up & Submission	May – Jun	Dissertation, poster, presentation

Success Criteria

- Working mobile prototype with core features.
- Usability score (SUS) > 70 from pilot users.
- Evidence of iterative design (sketches, prototypes).

Demo / Repository


 Scan for Live Demo

Supervisor: Mamoon Humayun • Module: Final Year Project • University of Roehampton
Contact: rowlandd1@roehampton.ac.uk

Figure 30: Project Proposal Poster

Appendix B: Project Management Tools

This appendix provides evidence of the project management tools and methodologies implemented and updated throughout the project's development lifecycle. Separate GitHub projects were created to track each component of the project: the artefact and the report.

Appendix B.1: Artefact Kanban Boards

The 'Artefact' GitHub project consisted of a kanban board in which tasks were written as user stories, with deliverables outlined as checklists within the description. Once a task was written, it would arrive in the backlog before moving to being ready. Once the task was in the stage of development it would transition to the in-progress tab. After all checklist items had been complete, the task would transition to the in review tab before arriving at its terminal state of 'Done'

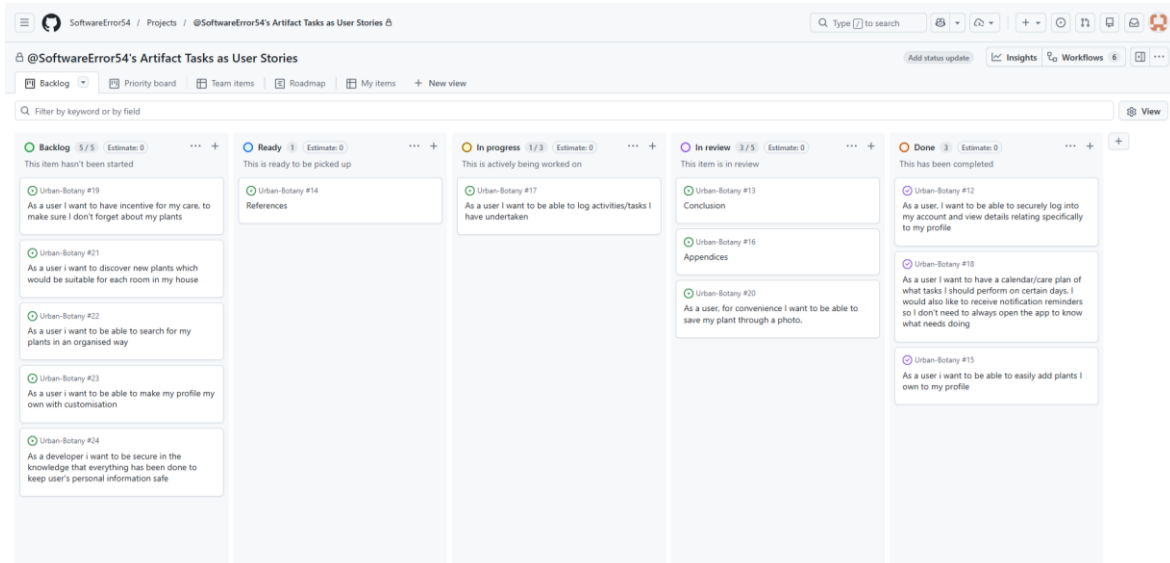


Figure 31: Artefact Kanban Board Overview

Figure 32 clearly displays the anatomy of a kanban board task, with its title being a 'User Story', framing the problem from the perspective of a real human issue, this improves the framing of the situation by applying subtext. Within the description, you can see the interactive checklist where developers can tick off a deliverable upon completion. This improved monitoring the state of a problem throughout its in progress state. Finally, below the comment section is a log of any state changes, movements or updates as well as who performed them, for team projects this blame-style feature improves communication between team members, knowing who has the most up to date information regarding a task.

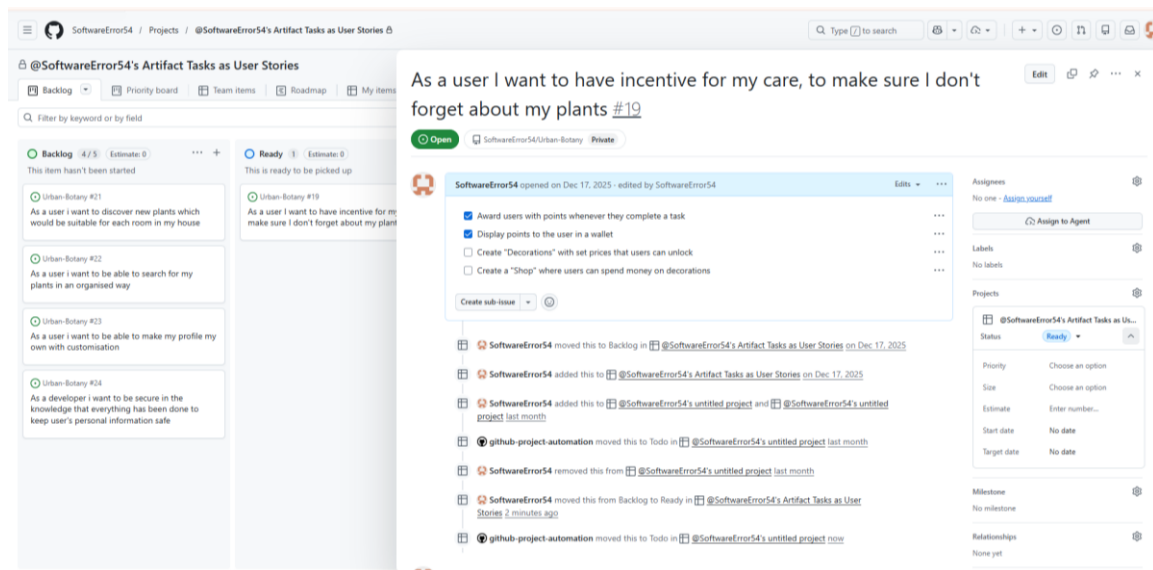


Figure 32: Task with User Story & Checklist

Appendix B.2: Report Gantt Charts

The report project within GitHub contained a list of headings for this report. It was used to plan project milestones and ensure tasks were completed in a timely manner, as seen in Figure 33.

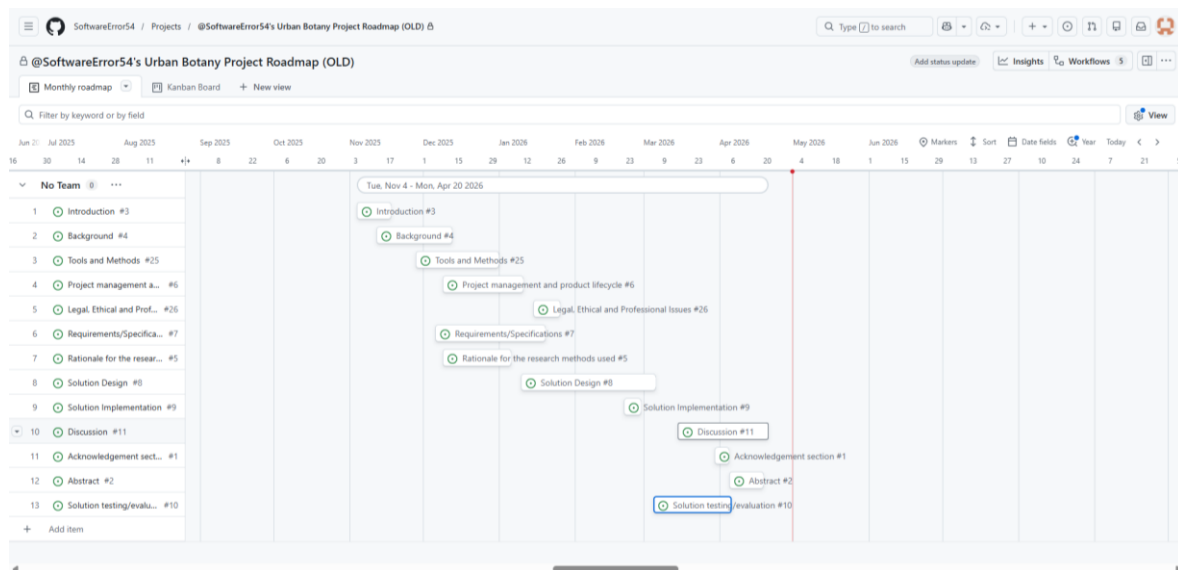


Figure 33: Report Gantt Chart

The report project did also feature a kanban board view of the project for improved transitioning of tasks through state as seen in Figure 3. However, the Gantt Chart view was most effective for maintaining deadlines.

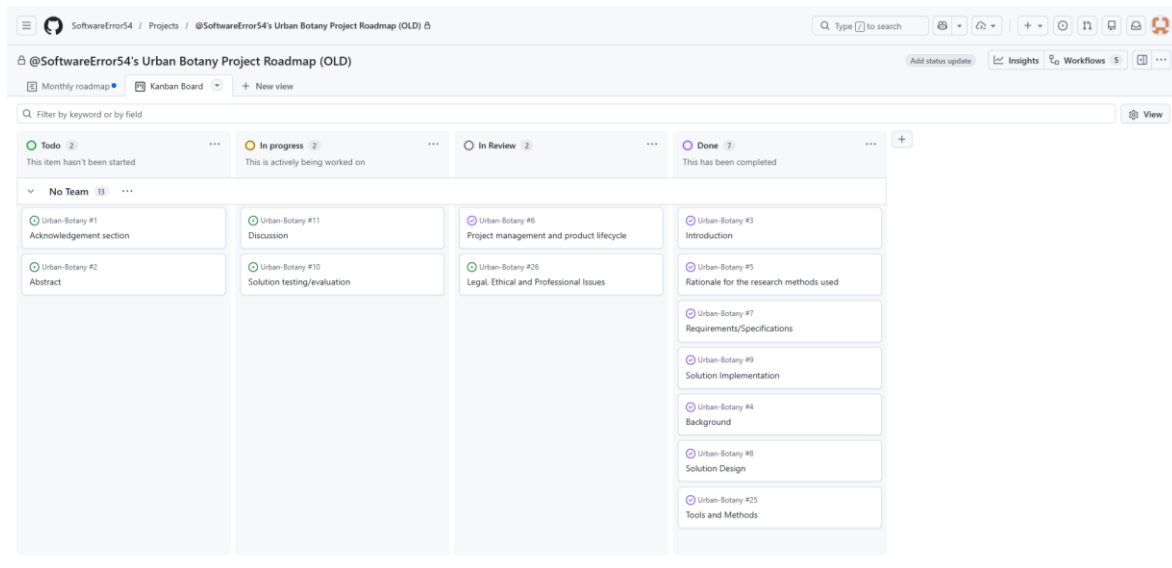


Figure 34: Kanban Board View of Report Tasks

Like the kanban boards, the Gantt Chart tasks featured an anatomy to help guide the production of written sections. Deliverables, and requirements were taken from the report brief and held within the tasks along with a rough plan of what needs to be written and in what order. The layout of a report task can be clearly seen in Figure 35.

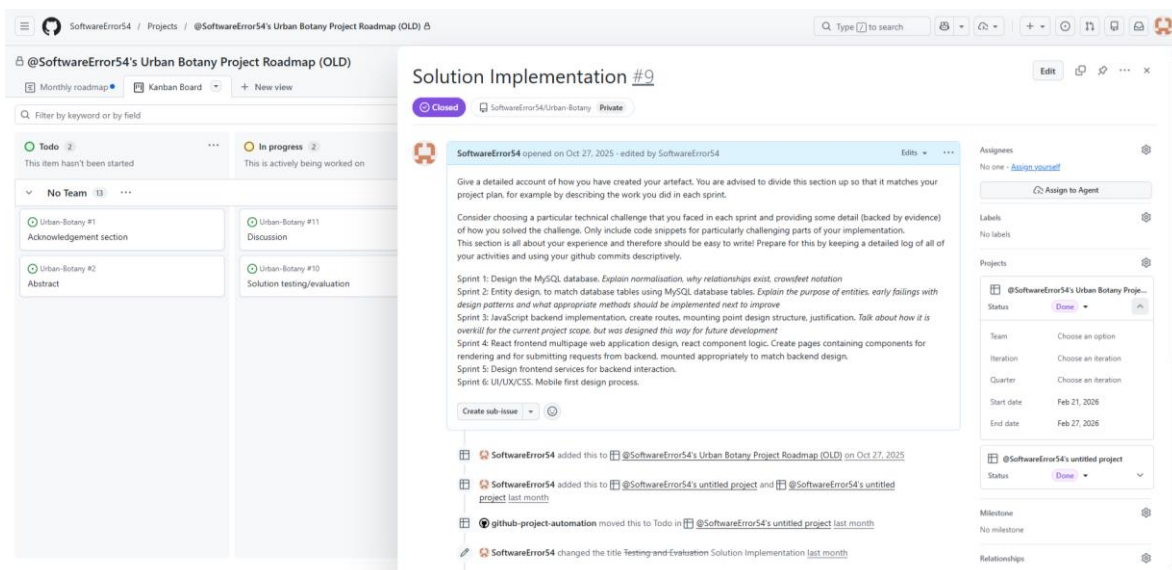


Figure 35: Report Task with Description

Appendix C: Technical Access and Artefact

The related artifact “Urban Botany” is in a public GitHub repository, below are the URL and QR code which can be used to access the project.

<https://github.com/SoftwareError54/Urban-Botany>



Figure 36: QR Code to Urban Botany Repository

Urban Botany is a full-stack web application that helps users living in urban dwellings care for their houseplants. Core features include plant identification via the PlantNet API, virtual room decoration, care task scheduling with reminders and a personal plant collection manager.

Artwork for the application has been hand drawn and currently resides within the application. Due to the limitation of team size, plant species and decorations are limited in this first version. Current species and decorations can be located within the public directory. The backend and frontend have been designed to allow for expansion of these species by assigning pointers to new image locations inside of the database and placing the new ‘.png’ files inside of the public directory in the relevant subdirectory.

Appendix C.1: Project Structure

Urban-Botany/

```

├── Backend/
|   ├── Controllers/    # Route handler logic
|   ├── Services/       # Business logic layer
|   ├── Repositories/   # Database query layer
|   ├── Entities/       # Data models
|   ├── Middleware/     # JWT auth middleware
|   ├── Routes/         # Express route definitions
|   └── config/db.js    # MySQL connection config
|
├── Frontend/
|   ├── src/
|   |   ├── Pages/      # Top-level page components
|   |   ├── PlantComponents/ # Plant-specific UI components
|   |   ├── RoomComponents/ # Room/scene UI components
|   |   ├── TaskComponents/ # Task/calendar UI components
|   |   └── services/    # API client and builder utilities
|   ├── public/         # Static assets (plant/room images, icons)
|   └── diagrams/        # Mermaid source files + generated PNGs
|
├── docker-compose.yaml
└── urbanbotany.sql      # Database schema + seed data

```

Above lies the project structure/architecture, to assist with any further development others would like to make to the project by forking the repository. The structure lays out and describes both hierarchy and function of all directories and some subdirectories. Frontend and Backend were separated to improve security in design, and allow for improved scaling.

Appendix C.2: Installation Guide

To setup this artifact on your local environment, the following steps should be followed. For consistency and reduce likelihood of issues, it is recommended that setup is taken using Docker.

1. Clone the repository:
git clone https://github.com/SoftwareError54/Urban-Botany.git
cd Urban-Botany
2. Create root environment variables:
MYSQL_ROOT_PASSWORD=your_root_password
MYSQL_DATABASE=urbanbotany
MYSQL_USER=ubuser
MYSQL_PASSWORD=your_db_password
3. Create Backend environment variables:
PORT=3000
MYSQL_HOST=mysql
MYSQL_PORT=3306
MYSQL_USER=ubuser
MYSQL_PASSWORD=your_db_password
MYSQL_DATABASE=urbanbotany
JWT_SECRET=your_jwt_secret_here
JWT_EXPIRES_IN=1h
PLANTNET_API_KEY=your_plantnet_api_key
4. Start all services:
In a new terminal, run: docker compose up --build
On subsequent builds, simply run: docker compose up
5. Access the application:

Table 8: Services and Corresponding URLs

Service	URL
Frontend	http://localhost:5173
Backend	http://localhost:3000
phpMyAdmin	http://localhost:8080

6. Stop services:
To stop the services in detached mode, in the terminal run: docker compose down
Alternatively press ctrl + c
To reset any mounted volumes, run: docker compose down -v

Appendix D: Risk Register

This appendix presents the visual representation of risks identified within the main body of the report. Risks were assessed on likelihood of an event occurring, the impact in the event of such risks coming to fruition, and the mitigation strategies that can be implemented to reduce either likelihood or impact.

Table 9: Risk Register

Risk	Likelihood	Impact	Mitigation Strategies
Data breach	Medium	High	Automated compliance scanners, GitHub Workflows, Penetration Testing, Third Party Compliance, Internal Audits
IP and API Licencing	Medium	Low	Acknowledge the use of PlantNet API, and stay up to date on compliance
Model Accuracy	Medium/High	Low	Practice transparency, informing the user of how decisions are made through thorough documentation, providing confidence scores where possible
Identification of Toxic Species	Medium	High	Where plants are flagged as toxic or have visually similar toxic counterparts, inform users of risk. Additionally, advise users against consumption of plants
Accessibility based reputational damage	Low	Low	Implement the use of tools such as NVDA, JAWS, VoiceOver, and WAVE to test and scan for accessibility concerns
Product failure	Medium	Medium	Implement the use of project management tools and processes such as kanban boards and Gantt charts
Engagement	Medium	Low	Implement monitoring of click-through-rates, A/B testing, average time spent on a page, login frequency and response to stimulus to monitor and limit engagement

Appendix E: Stakeholder Communication

Within this appendix is the questionnaire template and some of the responses provided by stakeholders. These responses were used to identify target groups and from those target groups common features were extracted to create the personas used within the planning stages of the artefact. Any names and personal details have been removed from the responses to maintain stakeholder confidentiality.

Appendix E.1: Persona Interview Template

Persona Interview Template

Table 10: Blank Persona Template

Do you have any house plants at home?

What is your favourite plant that you have?

How do you feel about your houseplants?

Do plants impact your impressions of your environment?

Where in your home do you keep your plants?

--

Talk me through your average week

--

Do you enjoy having house plants around?
--

--

Why do you think this is?

--

Do you struggle with caring for your plants?
--

--

What do you consider your biggest struggles when it comes to looking after plants?
--

--

With this in mind, what do you want to improve on?
--

--

What tools and processes do you feel would help you improve on these?

--

Appendix E.2: Questionnaire Response 1

Table 11: Persona Questionnaire 1

Do you have any house plants at home?

Yes, I have around 50 houseplants

What is your favourite plant that you have?

My favourite plant would probably be my Thai Constellation monstera named Princess
--

How do you feel about your houseplants?

I love my plants, having something to care for gives me a sense of responsibility

Do plants impact your impressions of your environment?
--

I like having a lot of green at home, I feel like plants add a nice touch and cleans the air at home. I also love to see them grow
--

Where in your home do you keep your plants?

We have our plants all over the apartment, wherever there is space on a windowsill or table with suitable light

Talk me through your average week

I have a very varied weekly routine, since I'm active in different student groups my weeks look different. Some days I come home late (around 8pm-ish) others I'm home all day. Since my partner is doing distance studies he usually cooks in the evening, so it's done when I get home. I do a lot of commuting to campus, around 2-3 hours per day.

Do you enjoy having house plants around?

See prior answers

Why do you think this is?

See prior answers

Do you struggle with caring for your plants?

I have learned to know what most of my plants need, but sometimes I do struggle with new ones, or fussy ones

What do you consider your biggest struggles when it comes to looking after plants?

Wrong soil mix, pests, overwatering, underwatering, too much sun, too little sun. I have a lot of dramatic plants.

What do you want to improve on?

I would like to improve on the quality of care I can provide to my plants

What tools and processes do you feel would help you improve on these?

A guide on soil mixes, how much/often to water based on light, soil, plant, and temperature

Appendix E.3: Questionnaire Response 2

Table 12: Persona Questionnaire 2

Do you have any house plants at home?

Currently has two plants

What is your favourite plant that you have?

Yes, the one in her room. Given as a gift for a birthday. A big leafy plant

How do you feel about your houseplants?

Plants are not real people, doesn't have a name for it. It makes her happy when she sees it

Do plants impact your impressions of your environment?

100% Believes that plants do wonders for mental health.

Where in your home do you keep your plants?

One in her bedroom and one in the downstairs bathroom

Talk me through your average week

Focusing on exercising, running, gym, working in hospitality while actively pursuing her future career. Enjoys reading and meeting up with friends.

Do you enjoy having house plants around?

One of life's simple pleasures

Why do you think this is?

See above

Do you struggle with caring for your plants?

Only if they require a lot of attention

What do you consider your biggest struggles when it comes to looking after plants?
--

They can't communicate their problems to me, I don't know what is wrong with them when they droop and wither
--

With this in mind, what do you want to improve on?
--

I'd like to better understand what is wrong with them when they start to die
--

What tools and processes do you feel would help you improve on these?

A watering can, an irrigation system at home. Better education in schools.
--

Appendix E.4: Questionnaire 3

Table 13: Persona Questionnaire 3

Do you have any house plants at home?

Yes, I have 11

What is your favourite plant that you have?

A snake plant

How do you feel about your houseplants?

They bring joy, peace of mind

Do plants impact your impressions of your environment?

Makes it feel fresher and cleaner.

Where in your home do you keep your plants?

There are 4 in the living room, the rest are scattered around bedrooms on windowsills

Talk me through your average week

Wake up, has a skin clinic working ad-hoc. Spends most of the time at university in lectures or lab helping. Goes on morning runs 3x per week. Enjoys reading. Spends a good amount of time at home.

Do you enjoy having house plants around?

See above

Why do you think this is?

See above

Do you struggle with caring for your plants?
Struggles to care for each plant.

What do you consider your biggest struggles when it comes to looking after plants?
Individual care based on their needs. Watering them in time.

With this in mind, what do you want to improve on?
Remembering to check on my plants

What tools and processes do you feel would help you improve on these?
Reminders to check on plants frequently. Knowing what each plant needs